Article

Graph Neural Network for Traffic Forecasting: The Research Progress

Weiwei Jiang [1,*], Jiayun Luo [2*], Miao He [3] and Weixi Gu [4,*]

[1] School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing, China;
[2] School of Computer Science and Engineering & China-Singapore International Joint Research Institute, Nanyang Technological University, 639798, Singapore;
[3] Yanqi Lake Beijing Institute of Mathematical Sciences and Applications, Beijing, China;
[4] China Academy of Industrial Internet, Beijing, China.
* Correspondence: jww@bupt.edu.cn; guweixi@china-caii.com

Abstract: Traffic forecasting has been considered as the basis for many intelligent transportation system (ITS) applications, including but not limited to trip planning, road traffic control and vehicle routing. Various forecasting methods have been proposed in the literature for Internet traffic prediction, including statistical methods, shallow machine learning algorithms and deep learning algorithms. More recently, graph neural networks (GNNs) emerge as the state-of-the-art solutions of traffic forecasting due to its natural fit to the graph-like traffic system. Therefore, this survey aims to present the research progress on graph neural networks for traffic forecasting and the research trends observed from the most recent studies. In addition, this survey summarizes the latest open source dataset and code resources for sharing with the research community. Finally, the research challenges and opportunities are proposed to inspire follow-up studies.

Keywords: Traffic Forecasting; Graph Neural Network; Graph Convolutional Network; Graph Attention Network.

1. Introduction

Traffic forecasting is fundamental to the construction of modern transportation infrastructures and intelligent transportation systems (ITSs). It has a wide range of applications such as in trip planning, road traffic control and vehicle routing [1–4]. As a crucial prerequisite for the operation of the community, traffic forecasting has drawn a great amount of attention from both academia and industry [5–8]. However, the traffic forecasting problem is not fully resolved because of the complex spatial and temporal dependencies of traffic activities. Furthermore, the development of the Internet of Things (IoT), the Internet of Vehicles (IoV) and artificial intelligence (AI) techniques [9] enable measuring and modeling more diverse traffic related features, which then allows the design of autonomous and efficient data-driven traffic forecasting methods [10]. In order to obtain a bird-eye view of the opportunities and challenges exist in traffic forecasting, we here summarize the up-to-date research progress in this vibrant field to facilitate future study.

The traffic forecasting problems can be categorized into different types based on the data formats utilized in the study, including time series data, grid data and graph data. Among these, the earliest and most general problem formulation is time series forecasting [11–14], where the historical data points are used as model input to predict future situations. Moreover, the time series forecasting problem can be divided into univariate and multivariate problems. For the univariate time series problems, only one traffic variable is considered, e.g., traffic flow or traffic speed. For multivariate time series problems, multiple traffic variables are considered simultaneously. Besides the univariate and multivariate setting, time series forecasting can also be formulated into single-step forecasting and multiple-step forecasting. In the single-step forecasting problem, only one

Citation: Weiwei J.; Jiayun L.; Miao H. and Weixi G. Graph Neural Network for Traffic Forecasting: The Research Progress. *ISPRS Int. J. Geo-Inf.* **2022**, *1*, 0. https://doi.org/

Received:
Accepted:
Published:

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.

data point in the next step is to be predicted. In the multiple-step forecasting problem, more than one value are to be predicted.

Although time series data is the most commonly used data format in the traffic-related studies, it is insufficient as the spatial dependency of traffic activities is not considered. To overcome this issue, two data formats are further developed, i.e., grid data and graph data. For traffic forecasting with grid data, at each time step, the traffic data are aggregated by some regularly divided regions in the city area being studied. Each regularly divided regions can be seen as a grid and by aggregating the corresponding traffic variables in each grid, we obtain an intensity map that can be displayed in image format, as shown in Figure 1. To utilize grid data for traffic forecasting, taking single-step forecasting as an example, the historical grid data, formulated as image frames, within a specific time sliding window are used as input features, and the frame in the next time step is the prediction target.

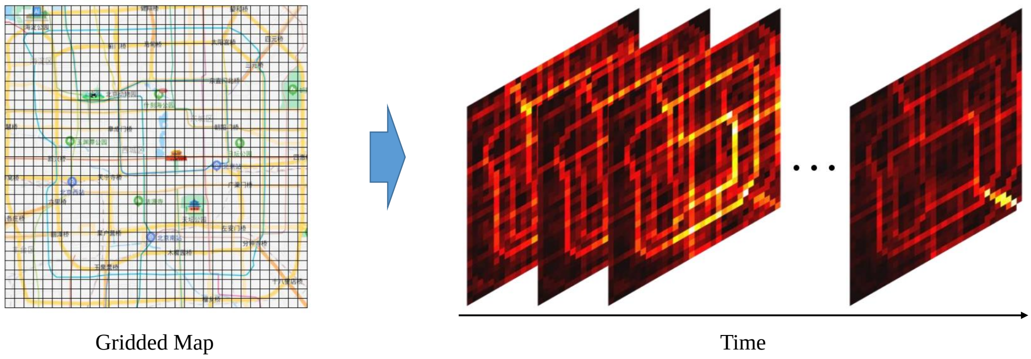

Figure 1. The grid-format traffic forecasting problem [15,16].

For graph-format traffic forecasting, the traffic data are aggregated by some locations or stations, which are considered as nodes in a traffic relation graph. The node features are the collected traffic variables, e.g., traffic flow or speed. The edges can model the road topological connection or the spatial distances among different nodes. Similarly, taking the single-step forecasting as an example, the historical graph data in a time sliding window are used as input features, and the graph in the next time step is the prediction target, as shown in Figure 2.

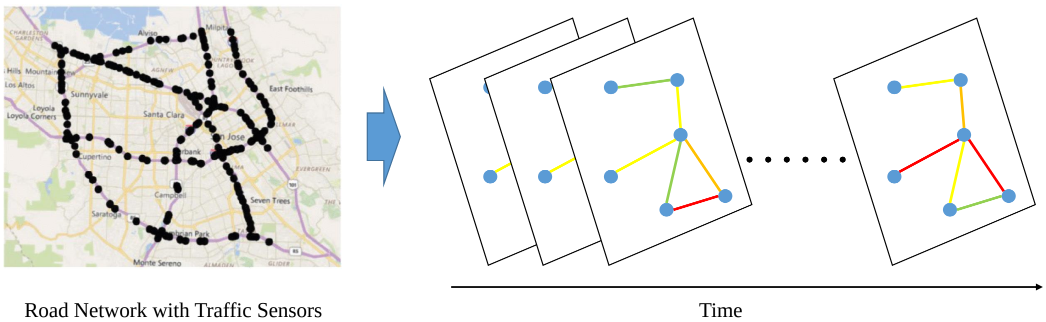

Figure 2. The graph-format traffic forecasting problem [17,18].

The existing traffic forecasting methods can also be assign into different categories according to models utilized, including statistical models, shallow machine learning models and deep learning models. Each of which has their respective application ranges and can fit different situations [19]. Statistical models are mainly linear models, e.g., autoregressive integrated moving average (ARIMA). These models are advantageous because of their low computation cost and good interpretability. However, their forecasting performance is inferior to machine learning and deep learning models, which are better at capturing nonlinear relationships. Shallow machine learning models are represented by tree-based models, e.g., decision trees and random forests [20]. They have been the first choices in the earlier studies until the recent adoption of the more autonomous and accurate deep learning models, represented by modern neural networks, e.g., convolutional neural networks (CNNs) [21] and recurrent neural networks (RNNs) [22].

These deep learning models has been proven effective for various forecasting problems in the financial, energy and communications sectors [23–33] and are being actively studied in traffic related domain. Among various deep learning algorithms, graph neural networks (GNNs) have become state-of-the-art solutions in traffic forecasting problems. GNNs leverage graph structures, which is commonly seen in traffic infrastructure, such as in road networks and subway systems. The interaction among nearby traffic sensors or stations can be effectively captured by GNNs, thus results in improved forecasting performance.

While there are already some surveys about deep learning for traffic forecasting problems, most of them are not GNN-focused, with only a few exceptions [34–41]. This study serves as an extension of the existing GNN-relevant surveys [42,43], summarizes the latest research progress in 2022 and aims to be the most updated reference manual for researchers in relevant fields. In this survey, a total of 120 journal papers and 30 conference papers published in 2022 are reviewed, all of which are selected from prestigious journals and conferences in transportation, computer science and multidisciplinary fields. Each paper is reviewed in a structured way, and the learned lessons are discussed to reveal research trends. Based on the surveyed studies, the newest open dataset and code resources are also collected and organized in lists. The existing research challenges are identified, and corresponding research opportunities are further proposed.

The contributions of this survey are summarized as follows:

- This survey summarizes the most updated researchers with the theme of graph neural networks for traffic forecasting.
- This survey provides the lists of the newest open dataset and code resources for the research community.
- This survey points out the existing research challenges and proposes corresponding research opportunities to inspire follow-up studies.

The rest of this paper is organized as follows. Section 2 is the literature review for the latest relevant studies as well as the discussion of recent research trends. Then, the lists of the newest open dataset and code resources for the research community are presented in Section 3. Finally, research challenges and opportunities when applying graph neural networks for traffic forecasting are discussed in Section 4 to inspire follow-up studies. The conclusion is drawn in Section 5.

2. Literature Review and Research Trends

The studies in this survey are all selected from prestigious journals and conferences in transportation, computer science and multidisciplinary fields. To share incremental knowledge and to avoid repetition with existing similar surveys [42,43], only those published in 2022 are selected and discussed in this section, with a total number of 120 journal papers and 30 conference papers. The source journals and conferences are listed in Table A2 and Table A3, with the number of papers counted. The reviewed studies are summarized in Table 1. For each study, the specific traffic problem, graph type, dataset, model component (especially the involved GNN structures) as well as the summary of the main content are discussed. More relevant studies are tracked and updated in our GitHub repository [1].

As in the introduction Section 1, we classify traffic forecasting problems from two angle of view, namely based on the data format, or based on the model used. Additionally, in Table 1, we provide another perspective, which is based on the transportation mode, e.g., road traffic, taxis, bikes, and subways. As indicated in Table 1, we find that across different traffic related studies, road traffic flow and speed prediction problems are still the most popular traffic forecasting problems. There are two possible reasons for this trend. The first

[1] https://github.com/jwwthu/GNN4Traffic

reason is that for road traffic forecasting problems, the open datasets and baseline models are more accessible, with well-processed steps and instructions, which save efforts for data collection and preprocessing. The second reason is that it is more intuitive to construct graphs for road network related problem, making it more natural to leverage GNNs for road traffic flow and speed prediction problems, thus more common to see under the scope of our survey.

As shown in the table, two graph types are listed in the graph column, namely, static graph and dynamic graph. In the early research stage, static graphs are widely used because of their convenience. However, researchers soon realize that static graphs are not enough to capture the network topology and traffic pattern changes. For example, the traffic flow measurements on the road segments and their correlations will change dynamically in space and time, which is beyond the modeling ability of static graphs. Then, dynamic graph is introduced. As indicated by the names, dynamic graphs are graphs that can evolve with new nodes or edges being added or deleted. Nevertheless, static graphs are still quite useful when the traffic infrastructure remains unchanged for the considered time period. Therefore, there is also researches utilize both dynamic and static graphs simultaneously. The static graph is used to model the static road network and the dynamic graph is used to consider the influence of dynamic traffic events and weather information.

Switching gear to the dataset used in the collected studies, most of them are open datasets with only a few exceptions. Among those open datasets, some have made great contributions for supporting relevant studies, with which different models can be tested and compared in a fair way, e.g., PeMS-BAY and METR-LA. However, it would also bring problems when the existing datasets are over-utilized and overfitted with GNN-based deep learning models and produce unreliable models for other traffic scenarios and datasets. To solve this potential risk, new datasets are collected and listed in Section 3 for further comparison. Moreover, interestingly, most of the surveyed studies use two or more datasets, which is a phenomenon whose benefits worth digging out and might provide insight with some comprehensive evaluation in the future.

For the model component part, the graph convolutional network (GCN) [44] and graph attention network (GAT) [45] are two dominate networks used. GCN is the pioneer work that migrates the concept of convolution operations from Euclidean image data to non-Euclidean graph data and has become extremely successful in the past few years. The basic idea of the GCN is to aggregate the features from the neighbors and then apply linear transformation on the aggregated features. GCN layers can be stacked k times to capture k-hop neighbor information. However, the GCN requires the whole graph structure for training, which consumes lots of computer memory. Under such scenario, GAT, based on the attention mechanism [46], is introduced as an alternative to GCN. The main distinction of GAT from GCN is the introduction of importance scores for different neighbors, based on the masked self-attention mechanism. Technical details about GCN and GAT are beyond the scope of this survey, which aims to identify the research trend, and can be found in relevant surveys [47,48]. Overall, the research trend remains to be developing more effective GCN or GAT variants, as can been foreseen, unless some fundamental breakthrough is made in the GNN research community.

Both the GCN and GAT are mainly used for capturing spatial dependencies. In order to capture the temporal dependencies, there are some classic models, e.g., temporal convolutional network (TCN), long short-term memory (LSTM) and gated recurrent unit (GRU). More recently, the attention-based model, i.e., Transformer, is proven effective for capturing long-term dependency in the time series [49]. Nonetheless, as indicated in Table 1, Transformer is only used in a few of the surveyed studies, and there is still much research space.

Table 1. Summary of the surveyed studies.

Study	Problem	Graph	Dataset	Model Component	Summary
[50]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	PeMS03, PeMS04, PeMS07, PeMS08, PeMS-BAY, METR-LA	GCN, TCN	Dual dynamic spatial-temporal graph convolution network (DDSTGCN) is featured by a dual graph structure of traffic flow graph and its dual hypergraph to reveal more complicated latent relations.
[51]	Road Traffic Flow	Dynamic Graph, Static Graph	PeMS04, PeMS08	TCN, GCN	Spatio-temporal adaptive graph convolutional network (STAGCN) is featured by the adaptive graph generation block to capture both the learnable long-time static road graph and the learnable short-time dynamic graph.
[52]	Road Traffic Speed, Regional Bike Flow	Dynamic Graph	PeMS-BAY, METR-LA, BikeNYC	GAT, TCN	JointGraph is featured by a network reconstructor to reconstruct the traffic graph and the ability to handle a multi-dataset joint training task.
[53]	Metro Traffic Flow	Dynamic Graph, Static Graph	BJMF15	GCN, TCN	Knowledge graph representation learning and spatiotemporal graph neural network (KGR-STGNN) is featured by knowledge graph representation learning which better captures the influence of external factors.
[54]	Regional Traffic Flow	Static Graph	HaikouTaxi, ChengduTaxi	GCN, TCN	Multi-Attribute Graph Convolutional Network (MAGCN) is featured by the consideration for area attributes and a novel matrix of which values are the functional area-based origin-destination pairs.
[55]	Ride-hailing Demand	Static Graph	Ride-hailing datasets in Beijing and Shanghai	GCN	The proposed multi-linear relationship GCN is featured by multi-modal coordinated representation learning and spatial feature extraction from different modalities.
[56]	Road Traffic Flow	Static Graph	PeMS08, METR-LA	GCN, LSTM	Multi-View Bayesian spatio-temporal graph neural network (MVB-STNet) is featured by the Bayesian neural network layer for handling data uncertainty with sparse and noisy data.
[57]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD7, PeMSD8	GraphSAGE, GRU	Transferable federated inductive spatial-temporal graph neural network (T-ISTGNN) is featured by the capability of cross-area traffic state forecasting when preserving the privacy of source areas.
[58]	Regional Taxi Usage	Static Graph	TaxiNYC	GAT, GRU	Spatio-temporal heterogeneous graph attention network (STHAN) is featured by a spatio-temporal heterogeneous graph in which multiple spatial relationships and temporal relationships are modeled and meta-paths are used to depict compound spatial relationships.
[59]	Road Traffic Flow	Dynamic Graph, Static Graph	METR-LA, PEMS-BAY	GCN, GRU	Spatiotemporal prediction framework using high-order graph convolutional network (STHGCN) is featured by a dynamic adaptive spatial graph learning module to learn the high-order dependence.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[60]	Road Traffic Flow	Dynamic Graph	PeMS04, PeMS08	GCN	The proposed CTVI+ framework is featured by a temporal self-attention mechanism and a multi-view graph neural network for learning temporal and spatial traffic patterns.
[61]	Origin-destination Demand	Dynamic Graph	TaxiNYC, BikeNYC, BikeDC	GAT, LSTM	Temporal graph Autoencoder (TGAE) is featured by a temporal network embedding framework that utilizes node representations in latent space to capture the temporal evolution of traffic networks.
[62]	Regional Ride-hailing Demand	Dynamic Graph	UberNYC, TaxiNYC	GAT, 1D-CNN, Transformer	Deep multi-view spatiotemporal virtual graph neural network (DMVST-VGNN) is featured by an integrated structure of GAT, 1D-CNN and Transformer networks.
[63]	Road Traffic Flow	Static Graph	Private Data	GCN, LSTM, GAN	Graph convolution and generative adversarial neural network [63] is featured by a GAN structure and parallel prediction ability for multiple steps.
[64]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	PeMS-Bay	GAT, GCN	Hierarchical mapping and interactive attention network (HMIAN) is featured by a hierarchical mapping structure for capturing functional zone relevance and long-distance dependence.
[65]	Road Traffic Flow	Static Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	The proposed forecasting framework is featured by the outlier detection strategy for a real-world IoV environment.
[66]	Metro Passenger Flow	Static Graph	Private Data	GCN, GRU	Spatial-temporal multi-graph convolutional wavelet network (ST-MGCWN) is featured by the graph wavelet convolution with multi-graph fusion.
[67]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, ConvLSTM	Multi-dimensional attention-based spatial-temporal network (MA-STN) is featured by the multi-dimensional attention mechanism to capture spatial and temporal patterns.
[68]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY	GCN, TCN	The proposed approach is featured by a multi-mode spatial-temporal convolution of mixed hop diffuse ordinary differential equation (MHODE).
[69]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN	The proposed approach is featured by a gated attention graph convolution model with multiple spatiotemporal channels.
[70]	Road Traffic Speed	Static Graph	PeMS-BAY, METR-LA	GCN, GRU	The proposed approach is featured by the combination of time classification and GCN models.
[71]	Road Traffic Flow, Bike Demand, Taxi Demand	Dynamic Graph, Static Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8, BikeNYC, TaxiNYC	GCN, GRU	Dual Graph Gated Recurrent Neural Network (DG^{2}RNN) is featured by a bidirectional GRU layer for learning temporal dependency and a spatial attention mechanism for learning spatial dependency.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[72]	Road Traffic Flow	Static Graph	Minnesota Department of Transportation Traffic Data	GCN, GRU	The proposed approach is featured by an attribute feature unit to fuse external factors into a spatiotemporal GCN.
[73]	Road Traffic Speed	Dynamic Graph	Seattle-Loop, METR-LA	GCN, GRU	Self-attention graph convolutional network with spatial, sub-spatial and temporal blocks (SAGCN-SST) captures the dynamic spatial dependency with a self-attention mechanism and is robust against traffic congestion and accidents.
[74]	Taxi Demand, Bike Demand	Dynamic Graph	TaxiNYC, BikeNYC	DCNN, Transformer	Dynamical spatial-temporal graph neural network (DSTGNN) is featured by the inhomogeneous Poisson process to model the changing demand process and the spatial-temporal embedding network to infer the intensity.
[75]	Ride-hailing demand	Dynamic Graph	TaxiNYC	GCN, GRU	Dynamic multi-graph convolutional network with generative adversarial network (DMGC-GAN) is featured by a multi-graph GCN module to learn from different dynamic OD graphs and a GAN structure to overcome the demand sparsity problem.
[76]	Road Traffic Speed	Static Graph	Private Data	GCN, GRU	The proposed approach is featured by the GAN structure for robust data-driven traffic modeling.
[77]	Road Traffic Flow, Road Traffic Speed	Static Graph	Seattle-Loop, PeMS-BAY	GAT, GAN	The proposed GAT-GAN framework is featured by the combination of first-order and high-order neighbors.
[78]	Road Traffic Flow	Dynamic Graph	PeMSD4, PeMSD8	GCN, CNN	Graph and attentive multi-path convolutional network (GAMCN) is featured by a novel GCN variant with road network graph embedding and a multi-path CNN module.
[79]	Road Traffic Accident	Dynamic Graph	NYC Open Data, PeMS-Bay	GCN	Multi-Attention Dynamic Graph Convolution Network (MADGCN) is featured by multiple attention mechanisms for capturing spatial and temporal influences.
[80]	Road Traffic Flow	Static Graph	Private Data	GCN, GAT	The proposed approach leverages DRL to integrate and improve GCN and GAT results.
[81]	Road Traffic Flow	Static Graph	PeMS (with 97 detectors)	GCN	The proposed approach is featured by the combination of GCN and six complex network properties.
[82]	Road Traffic Flow	Dynamic Graph	PeMSD7, PeMSD11	GCN	The proposed approach is featured by the GCN-based data imputation module and an adaptive approach of leveraging DRL for dynamic graph adjacency matrix generation.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[83]	Road Traffic Flow	Dynamic Graph	PeMSD4, PeMSD8	GCN	The proposed CRFAST-GCN is featured by the conditional random field (CRF) enhanced GCN to capture the semantic similarity globally.
[84]	Road Traffic Speed	Dynamic Graph	PeMSD8, METR-LA	TCN, GCN	A universal framework is proposed to transform the existing one-step-ahead models to multi-step-ahead models.
[85]	Road Traffic Speed	Static Graph	METR-LA, PEMS-BAY	GNN	The proposed approach is featured by a novel GNN layer with a location attention mechanism to aggregate traffic flow information from adjacent roads.
[86]	Road Traffic Speed	Dynamic Graph	METR-LA, PEMS-BAY	DCNN, TCN	Spatio-temporal sequence-to-sequence network (STSSN) is featured by an encoder-decoder structure with the joint modeling ability of spatial and temporal correlations.
[87]	Road Traffic Flow	Dynamic Graph	PeMS, Private Data	GNN, LSTM	Attentive attributed recurrent graph neural network (AARGNN) is featured by the modeling of both static and dynamic factors.
[88]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN	An adaptive graph learning algorithm (AdapGL) is proposed to learn the complex dependencies, and the model parameters are optimized with the expectation maximization algorithm.
[89]	Regional Traffic Demand	Dynamic Graph	UberNYC	GCN, CNN, TCN	Adaptive dual-view WaveNet (ADVW-Net) is featured by the WaveNet part to learn long-range temporal dependency.
[90]	Bike Demand, Taxi Demand	Static Graph	BikeNYC, TaxiNYC	GAT	Co-modal graph attention network (CMGAT) is featured by a multiple traffic graph-based spatial attention mechanism and a multiple time period-based temporal attention mechanism.
[91]	Road Traffic Speed	Dynamic Graph	METR-LA, PEMS-BAY	GCN, TCN	Adaptive spatio-temporal graph neural network (Ada-STNet) is featured by dedicated spatio-temporal convolution architecture and a two-stage training strategy.
[92]	Road Traffic Speed	Static Graph	PeMSD7, METR-LA, Seattle-Loop	GCN, Transformer	Attention based graph convolution network and Transformer (AGCN-T) is featured by the combination of GCN and temporal Transformer modules.
[93]	Road Traffic Speed	Dynamic Graph	PeMSD4, PeMSD8	GCN, ConvGRU	Attention encoder-decoder dual graph convolution model with time-series correlation (AED-DGCN-TSC) is featured by the combination of time series correlation analysis and deep learning modules.
[94]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	An improved dynamic Chebyshev GCN is proposed with a novel Laplacian matrix update method, the attention mechanism and a novel feature construction method.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[95]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, GLU	Causal gated low-pass graph convolution neural network (CGLGCN) is featured by the causal convolution gated linear unit with less computation time and GCN with self-designed low-pass filter.
[96]	Road Traffic Flow	Dynamic Graph	PeMSD4, PeMSD8	GAT	Attention-based spatiotemporal graph attention network (ASTGAT) is featured by multiple residual convolution and high-low feature concatenation.
[97]	Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY, PeMS-S	GCN	Attention-based dynamic spatial–temporal graph convolutional network (AD-STGCN) is featured by the combination of a dynamic adjustment module, a gated dilated convolution module and a spatial convolution module.
[98]	Road Traffic Flow	Static Graph	PeMS-LA, PeMS-BAY	GCN	Attention Based SpatioTemporal Graph Convolutional Network considering External Factors (ABSTGCN-EF) is featured by the combination of GCN and attention encoder network modules and the consideration of external factors.
[99]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, LSTM	Augmented multi-component recurrent graph convolutional network (AM-RGCN) is featured by a LSTM-based temporal correlation learner that incorporates one-dimensional convolution.
[100]	Road Traffic Speed	Static Graph	TaxiSZ	GCN, GRU	Bidirectional-Graph Recurrent Convolutional Network (Bi-GRCN) is featured by the combination of GCN and bidirectional GRU.
[101]	Road Traffic Speed	Static Graph	Private Data	GraphSAGE, LSTM	The proposed approach is featured by the combination of GraphSAGE, a global temporal block and the self-attention mechanism.
[102]	Regional Traffic Flow	Static Graph	Private Data	GCN, CNN	The proposed ConvGCN-RF is featured by a preprocessing-encoder-decoder framework and the combination of CNN, GCN and random forest modules.
[103]	Bus Demand	Static Graph	Private Data	GCN, LSTM	The proposed approach is featured by the combination of time-dependent geographically weighted regression and graph deep learning and the consideration of dynamic built environment influences.
[104]	Regional Crowd Flow	Static Graph	TaxiNYC, BikeNYC	GAT, CNN, LSTM	The proposed approach is featured by a semantic GAT module for learning dynamic inter-region correlations.
[105]	Road Traffic Speed	Static Graph	A new open data of Seoul, South Korea	GCN	Distance, direction, and positional relationship graph convolutional network (DDP-GCN) is featured by the consideration of three spatial dependencies.
[106]	Road Traffic Flow	Static Graph	PeMSD3, PeMSD7, Private Data	DGGP	The proposed approach is featured by novel deep graph Gaussian processes (DGGPs), which consist of the aggregation Gaussian process, the temporal convolutional Gaussian process and the Gaussian process with a linear kernel.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[107]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8, METR-LA, PeMS-BAY	GCN	Dynamic spatial-temporal adjacent graph convolutional network (DSTAGCN) is featured by the construction of the spatial-temporal graph and the integration of fuzzy systems and neural networks for uncertain relationship representation.
[108]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	PeMS-BAY, TaxiBJ, PeMSD4, PeMSD8	GCN, GRU	Dynamic spatial-temporal graph convolutional network (DSTGCN) is featured by a dynamic graph generation module with geographical proximity and spatial heterogeneity.
[109]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8, PeMS-SAN	GCN	The proposed approach is featured by a new temporal vector CNN module and a new dynamic correlation graph construction method.
[110]	Regional Travel Demand	Static Graph	TaxiNYC	GCN, GRU	The proposed approach is featured by the geographic similarity graph, functional similarity graph, and road similarity graph.
[111]	Road Traffic Speed	Dynamic Graph	PeMS-BAY, METR-LA	GCN, LSTM	The proposed EnGS-DGR model is featured by the ensemble learning of GCN, Seq2Seq and dynamic graph reconfiguration algorithms.
[112]	Road Traffic Speed	Dynamic Graph	PeMS-BAY, METR-LA	GCN, CNN, GRU	The embedded spatial-temporal network (ESTNet) combines multi-range GCN and 3D-CNN modules for modeling spatial-temporal dependencies.
[113]	Passenger OD Flow	Static Graph	Private Data	GCN, TCN	The proposed approach is featured by a novel sharing-stop network to model relationships between bus passengers and various mobility patterns.
[114]	Road Traffic Speed	Static Graph	Private Data	GCN, GRU	The proposed approach is featured by the incorporation of wavelet transform and usage of the Electronic Toll Collection (ETC) gantry transaction data.
[115]	Road Traffic Flow	Dynamic Graph	PeMSD4, PeMSD8	GAT	Fully dynamic self-attention spatio-temporal graph network (Fdsa-STG) is featured by the spatial GAT, temporal GAT and the fusion layers to extract recent, daily and weekly periodicity patterns.
[116]	Regional Traffic Flow	Dynamic Graph	TaxiNYC, TaxiBJ	GAT, GCN, LSTM	Federated deep learning based on the spatial-temporal long and short-term network (FedSTN) is featured by the recurrent long-term capture network module, attentive mechanism federated network module, and semantic capture network module to capture both spatial-temporal and semantic features.
[117]	Intersection Turning Traffic Flow	Static Graph	A new open data of Wuhan, China	GCN, GRU	The proposed approach is featured by the modeling of turning traffic flow with GCN and GRU.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[118]	Metro Ridership	Static Graph	Private Data	GCN, LSTM	Attention-weighted multi-view graph to sequence learning approach (AW-MV-G2S) is featured of learning spatial correlations from geographic distance, functional similarity and demand pattern views.
[119]	Regional Traffic Flow	Dynamic Graph	TaxiNYC, TaxiBJ, BikeNYC	GAT	The proposed approach is featured by the multi-resolution transformer network, GAT, and channel-aware recalibration residual network modules.
[120]	Regional Traffic Flow	Dynamic Graph	Private Car Trajectory Data, TaxiNYC	GCN, GRU, GAN	Multiple graph-based generative adversarial network (MG-GAN) is featured by the multi-graph dense convolutions with gated recurrent networks as the generative network and the attentive multi-graph convolutional network as the discriminative network.
[121]	Road Traffic Flow	Dynamic Graph	PeMSD3, METRA-LA	GAT	The proposed GDFormer is featured by a novel graph diffusing attention module to model the dynamically changing traffic flow.
[122]	Road Traffic Speed	Dynamic Graph	PeMS-BAY, NavInfo Beijing, NavInfo Shanghai	GAT	The proposed approach is featured by a novel data-driven graph construction method.
[123]	Road Traffic Flow	Dynamic Graph	Metro Interstate Traffic Volume Data Set	GAT, LSTM	Graph correlated attention recurrent neural network (GCAR) is featured by the combination of GAT, multi-level attention and parallel LSTM modules.
[124]	Road Traffic Speed	Dynamic Graph	Q-Traffic	GAT	Graph sequence neural network with an attention mechanism (GSeqAtt) is featured by two attention mechanisms to capture temporal correlations and graph structures.
[125]	Intersection Traffic Flow	Static Graph	Private Data	GCN	Spatial-temporal graph convolutional network (ST-GCN) is featured by an adjacent-similar algorithm and the ability to model both spatial and temporal dependencies of intersection traffic.
[126]	Regional Traffic Speed	Static Graph	Private Data	GCN, ConvLSTM	The proposed HDL4STP model is featured by the combination of GCN, ConvLSTM and fusion layers.
[127]	Road Traffic Flow	Dynamic Graph, Static Graph	PeMSD4, PeMSD8	GCN, LSTM	Improved graph convolution res-recurrent network (IGCRRN) is featured by the combination of the origin graph matrix and the data-generated embedding node matrix for spatial dependency.
[128]	Bike Flow	Static Graph	Private Data	Relation Graph Network	The proposed approach is featured by a generalized attention mechanism to extract block features and make cross-city predictions.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[129]	Subway Demand, Ride-hailing Demand	Static Graph	SubwayNYC, TaxiNYC	GCN	Multi-relational spatiotemporal graph neural network (ST-MRGNN) is featured by the multi-modal demand prediction ability with multi-relational GNNs.
[130]	Road Traffic Flow	Static Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8, PeMS-BAY	GAT, TCN	Multi-relational synchronous graph attention network (MS-GAT) considers multi-aspect traffic data couplings and learns channel, temporal, and spatial relations with GATs.
[131]	Road Traffic Speed	Dynamic Graph	Private Data	GAT, CNN	The proposed HA-STGN model considers spatial-temporal heterogeneous features and contains a dynamic graph module, a time-sensitive attention mechanism and an adaptive fusion module.
[132]	Road Traffic Flow	Static Flow	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	Adaptive graph cross strided convolution network (AGCSCN) is featured by temporal feature extraction with a cross strided convolution network and spatial feature extraction with an adaptive GCN.
[133]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, LSTM	Long short-term memory embedded graph convolution network (LST-GCN) is featured by the LSTM embedding into GCNs.
[134]	Road Traffic Speed	Dynamic Graph	DidiChengdu, METR-LA	GCN, TCN	Spatiotemporal adaptive gated graph convolution network (STAG-GCN) is featured by the combination of self-attention TCN, mix-hop adaptive gated GCN and fusion layers.
[135]	Road Traffic Flow	Static Graph	PEMS03, PEMS04, PEMS07, PEMS08	GCN, LSTM	Memory attention enhanced graph convolution long short-term memory network (MAEGCLSTM) is featured by the combination of memory attention mechanism and LSTM.
[136]	Road Traffic Speed	Dynamic Graph, Static Graph	PeMS-BAY	GCN, TCN	Multi-stage spatio-temporal fusion diffusion graph convolutional network (MFDGCN) is featured by multiple static and dynamic spatio-temporal association graphs.
[137]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, TCN	Multi-head self-attention spatiotemporal graph convolutional network (MSAS-GCN) is featured by the combination of GCN, TCN and the multi-head self-attention mechanism.
[138]	Metro Passenger Flow	Static Graph	HZMF2019	GCN, GAT, CNN	Multi-time multi-graph neural network (MTMGNN) is featured by the combination of Gated CNN, GCN and GAT modules with multiple graphs.
[139]	Road Traffic Speed	Static Graph	METR-LA	GCN, TCN	Gated temporal graph convolution network (GT-GCN) is featured by the multi-step ahead prediction ability with GCN and gated TCN modules.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[140]	Regional Ride-hailing Demand	Static Graph	Private Data	GCN, LSTM	Multigraph aggregation spatiotemporal graph convolutional network (MAST-GCN) is featured by a novel graph aggregation method.
[141]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD7, PeMSD8	GCN	The proposed approach is featured by the multi-scale traffic prediction ability with cross-scale GCN and temporal networks.
[142]	Metro Passenger Flow	Dynamic Graph	Private Data	GCN, GRU	The proposed approach is featured by the proposal of multi-featured spatial-temporal dynamic multi-graph convolutional networks for spatial and temporal connections.
[143]	Road Traffic Speed	Static Graph	Q-Traffic, Private Data	GCN, LSTM	The proposed approach is featured by the multi-fold correlation attention network to model dynamic correlations.
[144]	Regional Traffic Flow	Dynamic Graph	TaxiNYC, BikeNYC	GCN, GRU	Multi-mode dynamic residual graph convolution network (MDRGCN) is featured by the multi-mode dynamic GCN, GRU and residual modules for learning cross-mode relationships.
[145]	Metro Passenger Flow	Static Graph	Private Data	GCN, GAT, LSTM	Temporal graph attention convolutional neural network model (TGACN) is featured by a multi-graph generation method and a new spatio-temporal feature fusion method.
[146]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY	GCN, Transformer	Multi-view spatial–temporal graph neural network (MVST-GNN) is featured by multi-view Transformer and GCN modules.
[147]	Metro Flow, Bus Flow	Static Graph	Private Data	GCN	Multi-task hypergraph convolutional neural network (MT-HGCN) models the correlation between different tasks with a feature compress unit.
[148]	Regional Traffic Flow	Static Graph	TaxiSZ	GCN, GRU	The proposed TmS-GCN model is featured by the combination of GCN and GRU modules.
[149]	Road Traffic Flow, Road Traffic Speed	Static Graph	Private Data	GCN, LSTM	The proposed method is featured by the Seq2seq GCN-LSTM framework and the usage of connected probe vehicle data.
[150]	Bus Passenger Flow	Static Graph	Private Data	GCN, LSTM	The proposed method is featured by the bus network graph construction method and the combination of GCN and LSTM modules.
[151]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN	The proposed approach is featured by the combination of graph deep learning and federated learning.
[152]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD7	GCN, GRU	Spatial-temporal attention graph convolution network on edge cloud (STAGCN-EC) is featured by the edge training approach and the deep learning modules designed for low computational power devices.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[153]	Road Traffic Flow	Static Graph	PEMSD3, PEMSD4, PEMSD7, PEMSD8	GCN, TCN	The proposed approach is featured by two semantic adjacency matrices and a dynamic aggregation method.
[154]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY	GCN, GRU	Spatial-temporal upsampling graph convolutional network (STUGCN) is featured by a novel upsampling method with virtual nodes to model the global spatial-temporal correlations.
[155]	Regional Passenger Demand	Static Graph	DidiCD, TaxiNYC	GAT, ConvGRU	The proposed approach is featured by the combination of GAT and ConvGRU modules.
[156]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, CNN	The proposed STGMN model is featured by the combination of 1D-CNN with channel attention and interpretable multi-graph GCN modules.
[157]	Metro Passenger Flow	Dynamic Graph	Private Data	GCN, TrellisNet	The proposed STP-TrellisNets+ incorporates TrellisNet with graph convolution in multi-step traffic prediction for the first time.
[158]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD8	GCN, TCN	Spatial-temporal global semantic graph attention convolution network (STS-GAN) is featured by the usage of global geographic contextual information for urban flow prediction.
[159]	Road Traffic Flow	Static Graph	PeMSD4	GAT, GLU	Spatio-temporal multi-head graph attention network (ST-MGAT) is featured by the combination of GAT and GLU structures.
[160]	Taxi Demand	Static Graph	Private Data	MPNN	The proposed approach is featured by multimodal message passing and attention mechanisms.
[161]	Road Traffic Congestion	Static Graph	PeMSD4	GAT	The proposed TCP-BAST is featured by the bilateral alternation modules with GAT, a multi-head masked attention network, and temporal and spatial embedding.
[162]	Road Traffic Flow, Road Traffic Speed, Road Travel Time	Static Graph	TaxiBJ, META-LA, PeMS-BAY, PeMSD4, PeMSD8	GCN, GAN, GRU	The proposed approach is featured by the combination of multi-graph GCN and GAN structures.
[163]	Road Traffic Flow	Dynamic Graph	PeMSD4, PeMSD7	TCN, GCN	The proposed framework is featured by the combination of dilated TCN, multi-view GCN, and masked multi-head attention modules.
[164]	Road Traffic Speed	Dynamic Graph, Static Graph	METR-LA, PeMS-BAY	GCN, GRU	Time-evolving graph convolutional recurrent network (TEGCRN) is featured by the combination of time-evolving and predefined graphs.
[165]	Road Traffic Speed	Static Graph	Seattle-Loop, TaxiSZ	GCN, GRU, GAN	The proposed approach is featured by the combination of GCN and GAN with output distribution constraints.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[165]	Road Traffic Speed	Static Graph	TaxiSZ, METR-LA, PeMS-BAY	MPNN, GRU	The proposed approach is featured by a combination of bidirectional message passing, GRU and self-attention mechanisms.
[166]	Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY	GAT, TCN	The proposed TransGAT model is featured by the attention-based node embedding algorithm and the gated TCN module.
[167]	Regional Ride-hailing Demand	Static Graph	DidiHaikou, Taxi-Wuhan	GCN, LSTM	Multi-view deep spatio-temporal network (MVDSTN) is featured by the combination of both traffic and semantic views.
[168]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY, PeMSD4, PeMSD7	Transformer	Adaptive graph spatial-temporal Transformer network (ASTTN) is featured by adaptive spatial-temporal graph modeling with local multi-head self-attention.
[169]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY, PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN, TCN	The proposed approach is featured by the neural architecture search framework for GNN and CNN modules.
[170]	Road Traffic Speed	Static Graph	METR-LA, PeMSD4	GCN, TCN	spatial-temporal channel-attention based graph convolutional network (STCAGCN) is featured by stacked dilated convolution for long sequence modeling.
[171]	Road Traffic Flow	Dynamic Graph, Static Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	The proposed approach is featured by a cascading structure to enhance interaction and capture heterogeneity.
[172]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY, PeMS-M, PeMSD4, PeMSD8	GAT	Spatio-temporal graph attention network (ST-GAT) is featured by the individual spatio-temporal graph for modeling individual dependencies.
[173]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY	GCN, TCN	The proposed approach is featured by a novel residual estimation module.
[174]	Bike Demand	Dynamic Graph	BikeChicago, BikeLA	GNN	The approach approach is featured by a novel graph generator and GNN with flow-based and attention-based aggregators.
[175]	Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-Bay, TaxiSZ	GCN	The proposed approach is featured by the decomposition of seasonal static and acyclic dynamic components for traffic prediction.
[176]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7	GCN, GRU	AdaBoost spatio-temporal network (Ada-STNet) is featured by the boosting approach of stacking base models.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[177]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY, PeMSD4, PeMSD8	GCN, GRU	Decoupled Dynamic Spatial-Temporal Graph Neural Network (D^2STGNN) is featured by a decoupled spatial-temporal framework and a dynamic graph learning module.
[178]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN, GTU	Dynamic spatial-temporal aware graph neural network (DSTAGNN) is featured by a new dynamic spatial-temporal aware graph and a novel GNN structure.
[179]	Road Traffic Flow	Static Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GNN	The proposed approach is featured by first-order gradient supervision (FOGS) which uses first-order gradients for training the prediction model.
[180]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	Spatio-temporal graph neural controlled differential equation (STG-NCDE) is featured by the incorporation of neural controlled differential equations in traffic forecasting for the first time.
[181]	Road Traffic Flow	Static Graph	PeMSD4, PeMSD7	GNN	This study proposes a communication-efficient federated learning approach for graph-based traffic forecasting.
[182]	Road Traffic Flow, Traffic Demand	Static Graph	PeMSD3, PeMSD8, BikeNYC, TaxiNYC	GCN, MSDR	The proposed approach is based on a graph-based multi-step dependency relation (MSDR) model with the ability to learn from multiple historical time steps.
[183]	Road Traffic Speed	Static Graph	DidiShenzhen, DidiChengdu, PeMS-BAY, METR-LA	GCN	The proposed ST-GFSL framework is featured by the combination of spatio-temporal traffic prediction with few-shot learning and cross-city knowledge transfer.
[184]	Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY	GAT, TCN	The proposed approach is featured by the semantic closeness relationship and the traffic dynamics.
[185]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	METR-LA, PeMS-BAY, PeMSD4	GNN	The proposed approach enhances the performance of spatio-temporal GNNs with a pre-training model trained with very long-term history data.
[186]	Road Traffic Flow	Dynamic Graph, Static Graph	PeMSD4, PeMSD8	GCN, GRU	Regularized graph structure learning (RGSL) is featured by the embedding-based implicit dense similarity matrix, regularized graph generation method, and Laplacian matrix mixed-up module to fuse the graphs.
[187]	Road Traffic Flow, Road Traffic Speed	Dynamic Graph	PeMSD8, METR-LA	GCN, TCN	Spatio-temporal latent graph structure learning network (ST-LGSL) is featured by a MLP-kNN based graph generator and the combination of diffusion graph convolutions and gated TCN modules.

Continued on next page

Table 1 – continued from previous page

Study	Problem	Graph	Dataset	Model Component	Summary
[188]	Road Traffic Flow	Static Graph	Private Data	GCN, GRU	Spatio-temporal differential equation network (STDEN) is featured by the combination of data-driven and physics-driven approaches and the differential equation network model for modeling the spatio-temporal dynamic process.
[189]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD8	GCN, CNN, GRU	Time-aware multipersistence spatio-supra graph convolutional network (TAMP-S2GCNets) is featured by the introduction of a time-aware multipersistence Euler-Poincaré surface and a supragraph convolution model for intra- and inter-dependencies.
[190]	Road Traffic Flow	Static Graph	PeMSD8	GCN, GRU, GLU	Two-stage stacked graph convolution network (ED2GCN) is featured by the stacking of GCN, GLU and attention mechanism.
[191]	OD Travel Demand	Static Graph	TaxiChicago, TaxiNYC	DCNN, TCN	Spatial-temporal zero-inflated negative binomial graph neural network (STZ-INBGNN) is featured by the uncertainty quantification of the sparse travel demand with diffusion and temporal convolution networks.
[192]	Road Traffic Speed	Static Graph	NAVER-Seoul, METR-LA	GCN	Pattern-matching memory network (PM-MemNet) is featured by a novel key-value memory structure and the pattern-matching framework.
[193]	Regional Traffic Flow	Static Graph	NeurIPS Traffic4Cast Challenge Data	GNN	The proposed approach is featured by the combination of U-Net with graph learning.
[194]	Road Traffic Speed	Static Graph	PeMS-BAY, METR-LA	GCN, CNN	The proposed approach is featured by the mix-hop GCN and stacked temporal attention mechanism.
[195]	Road Traffic Flow	Dynamic Graph	PeMSD3, PeMSD4, PeMSD7, PeMSD8	GCN	The proposed approach is featured by the graph construction method for cross-time and cross-space correlations.
[196]	Road Traffic Speed	Static Graph	METR-LA, PeMS-BAY	GCN, GRU	The proposed approach is featured by a novel local context aware spatial attention mechanism.
[197]	Road Traffic Speed	Dynamic Graph	PeMS-BAY, Private Data	GCN	The proposed approach is featured by the combination of GCN and attention for multi-dimensional information aggregation.

3. New Dataset and Code Resources

This section contributes the lists of the newest open dataset and code resources for the research community.

3.1. New Datasets

Open datasets are fundamental to the evaluation and comparison of different forecasting models. As discussed in Section 2, some open datasets have been widely used in the surveyed studies, e.g., METR-LA, PeMS and NYC Open Data. Despite the availability of these datasets, developing new datasets are still beneficial due to the following two reasons. The first reason is the risk of overfitting of deep learning models on existing datasets, especially those with a relatively small volume compared with the datasets in the other domains, e.g., large image and natural language corpus collections. The second reason is that, as transportation facilities keep changing, the dataset shift problem could occur for models trained on dataset collected years ago. The traffic patterns in the historical training data could be totally different from those in the newly collected test data, and the performance of trained deep learning models degrades greatly in unseen situations. Therefore, here we updated the community with the new publicly available traffic datasets in Table 2 to facilitate future study as well as to encourage constant renewing of high quality traffic dataset.

Table 2. The list of new open traffic datasets.

Study	Traffic Attributes	Spatial Range	Temporal Range	Download Link
[105]	Aggregated Taxi Speed	Seoul, South Korea	April 1, 2018 to April 30, 2018	https://github.com/SNU-DRL/ddpgcn-dataset
[117]	Aggregated Taxi Flow	Wuhan, China	July 1, 2015 to July 28, 2015	http://ggssc.whu.edu.cn/ggsscAssets/download/AttentionModel/code_and_data.zip
HZMF2019 [138]	Aggregated Metro Passenger Flow	Hangzhou, China	January 1st to January 25th, 2019	https://github.com/lixus7/MTMGNN
TaxiBJ21 [16]	Aggregated Taxi Flow	Beijing, China	November 2012, November 2014, and November 2015	https://github.com/jwwthu/DL4Traffic/tree/main/TaxiBJ21
[198]	Aggregated Traffic Flow	Beijing, China	June 1, 2009 to July 15, 2009	https://github.com/gao0628/Dataset
[199]	Aggregated Traffic Flow	Six Crosses in the Urban Area	56 Days	https://zenodo.org/record/3653880#.Y20cPHZBzT6
XiAn Road Traffic [200]	Aggregated Traffic Flow, Weather Data	Xi'an, China	August 1, 2019 to September 30, 2019	https://github.com/FIGHTINGithub/Xi-an-Road-Traffic-Data
[201]	Aggregated Traffic Flow	Aveiro, Portugal	2019, 2020, and 2021	https://figshare.com/s/d324f5be912e7f7a0d21
[202]	Aggregated Taxi and Bike Trips	New York City, USA	2019, 2020	https://github.com/Evens1sen/Deep-NYC-Taxi-Bike
[203]	Aggregated Taxi and Bike Trips	Chicago, USA	2013 to 2020	https://github.com/iipr/mobility-demand
[204]	Citywide Crowd Flow	Tokyo and Osaka	April 1, 2017 to July 9, 2017	https://github.com/deepkashiwa20/DeepCrowd

3.2. New Code Resources

Open code resources facilitate the replication of the published results and migration of the proposed models to new problems. We summarized here the publicly available new code resources in Table 3, with the implementation framework listed, i.e., TensorFlow [2] or

[2] https://www.tensorflow.org

PyTorch [3]. An observation from Table 3 is that PyTorch is becoming more popular than TensorFlow for developing new graph neural network models in traffic forecasting studies.

Table 3. The list of new open code resources.

Study	Framework	Link
DDSTGCN [50]	PyTorch	https://github.com/j1o2h3n/DDSTGCN
STAGCN [51]	PyTorch	https://github.com/QiweiMa-LL/STAGCN
CTVI+ [60]	PyTorch	https://github.com/dsj96/TKDD
TGAE [61]	PyTorch	https://github.com/wangqiang-codes/TGAE
GAMCN [78]	TensorFlow	https://github.com/alvinzhaowei/GAMCN
MADGCN [79]	TensorFlow, PyTorch	https://github.com/wumingyao/MADGCN
AdapGL [88]	PyTorch	https://github.com/goaheand/AdapGL-pytorch
Ada-STNet [91]	PyTorch	https://github.com/LiuZH-19/Ada-STNet
AM-RGCN [99]	PyTorch	https://github.com/ILoveStudying/AM-RGCN
DDP-GCN [105]	TensorFlow	https://github.com/SNU-DRL/DDP-GCN
GDFormer [121]	PyTorch	https://github.com/dublinsky/GDFormer
ST-GCN [125]	TensorFlow	https://github.com/Wautumn/ST-GCN
MTMGNN [138]	PyTorch	https://github.com/lixus7/MTMGNN2
TmS-GCN [148]	PyTorch	https://github.com/Joker-L0912/Tms-GCN-Py
STUGCN [153]	PyTorch	https://github.com/zsongsong/stugcn
D^2STGNN [177]	PyTorch	https://github.com/zezhishao/D2STGNN
DSTAGNN [178]	PyTorch	https://github.com/SYLan2019/DSTAGNN
FOGS [179]	PyTorch	https://github.com/kevin-xuan/FOGS
STG-NCDE [180]	PyTorch	https://github.com/jeongwhanchoi/STG-NCDE
ST-GFSL [183]	PyTorch	https://github.com/RobinLu1209/ST-GFSL
STEP [185]	PyTorch	https://github.com/zezhishao/STEP
RGSL [186]	PyTorch	https://github.com/alipay/RGSL
STDEN [188]	PyTorch	https://github.com/Echo-Ji/STDEN
TAMP-S2GCNets [189]	PyTorch	https://github.com/tamps2gcnets/TAMP_S2GCNets
PM-MemNet [192]	PyTorch	https://github.com/HyunWookL/PM-MemNet
[193]	TensorFlow	https://github.com/LucaHermes/graph-UNet-traffic-prediction
TraverseNet [205]	PyTorch	https://github.com/nnzhan/TraverseNet

4. Research Challenges and Opportunities

This section discusses the research challenges and opportunities when applying GNNs to the problem of traffic forecasting, so as to inspire follow-up studies.

4.1. Research Challenges

Several challenges can be observed from surveyed studies.

The first challenge is to boost traffic training data quality. Several data quality related problems could appear when leveraging graph neural networks. First, the cold-start problem [128] could occurs when the developed GNN models are to be used in new areas or stations. Second, the construction cost for high quality datasets is high as the data collection process can be time and money consuming. It would be even more difficult to collect a comprehensive dataset due to the infrequency of extreme or emergent traffic event, e.g., traffic congestion and accidents. Third, data privacy is also non-negligible if we want to create more comprehensive datasets, as most of the existing traffic datasets are collected from public transportation modes (e.g., taxis and shared bikes) or road sensors, instead of from private vehicles [206].

The second challenge is the diverse and complicated graph structure that exist in the real world traffic infrastructure. Most of the surveyed studies consider only dense graphs, e.g., in the city center areas or closely-connected highways, when traffic activities are active. However, the complete traffic graph of the city could be sparse where some nodes have

[3] https://pytorch.org/

no or few connections with other nodes. This real world condition has not received full attention in the surveyed studies. Another limitation of the surveyed studies is that the graphs considered are relatively small, e.g., less than 1,000 nodes. For example, the most popular PeMS datasets are a collection of subsets from a large dataset, which is collected from more than 40,000 individual detectors distributed in a much wider geographical area, as the size of the original dataset goes beyond the computing abilities of some research groups.

The third challenge is model robustness. Deep learning models have long been criticized for their black-box nature with no or little interpretation paired with the prediction result. This black-box problem exists for graph neural networks as well, and there are not much systematic methods for interpreting GNN under the traffic forecasting setting. Many abnormal situations or outliers in the data are removed in the processing step or do not appear in the training dataset. The performance of the trained GNN models would degrade when meeting these abnormal situations in the test or deployment stages, which could lead to large deviation in the model prediction. Considering such risks, it is important to enhance model robustness and interpretation ability to boost user confidence on the models.

4.2. Research Opportunities

To inspire future studies, some promising research opportunities are discussed, with the purposes of addressing the above-mentioned challenges and inspire future researches.

The first research opportunity is the introduction of traffic simulation tools for creating unseen and complex situations as training data. Two specific approaches can be further investigated, i.e., model-driven and data-driven approaches [207]. The model-driven approach is based on macroscopic or microscopic traffic simulators, in which the macroscopic tools focus on the high-level deterministic relationships of the flow, speed, and density of the traffic stream, while the microscopic tools focus on the individual details. On the other hand, the data-driven approach does not depend on traffic domain knowledge and creates more data samples from existing approaches, e.g., generative adversarial network (GAN)-based studies [75–77,120,162].

The second research opportunity is to introduce new learning schemes into the traffic forecasting problem, e.g., transfer learning, meta learning and federated learning. Transfer learning has been proven effective for transferring cross-city knowledge, which would be useful for dealing with the cold-start problem in the new city [183]. Furthermore, meta learning is proven useful for constructing new graph structures with effective structure-aware learning during the cross-city knowledge transfer process. Privacy preserving schemes are further proposed to be combined with transfer learning, protecting the sensitive information from the source domain [57]. Federated learning is another effective learning approach for maintaining data privacy while training effective deep learning models [151, 181].

The third research opportunity is the combination of knowledge graphs in different road conditions or transport modes, to build connections among them [53]. More external data can be used when constructing traffic knowledge graphs, e.g., the activity calendar from social media for potential traffic demands. Additionally, the knowledge among different transportation modes, e.g., the interchange hubs, would be useful for multi-modal prediction [129,144,147].

The fourth research opportunity is the distributed learning approach for training large-scale graph neural networks for traffic forecasting [208,209]. Distributed training of graph neural networks would be necessary when the application scale of GNNs for traffic forecasting expanded to much larger graphs. In those cases, the improvement of training and run-time efficiency would be more beneficial and significant. Another similar idea is to leverage cloud computing for model training and edge computing for run-time inference [152,210] to accelerate the distributed training and inference processes.

The fifth research opportunity is the Bayesian learning approach for uncertainty quantification. Bayesian neural networks have been proven effective for handling the data uncertainty caused by noise data or missing data in road traffic flow prediction [56]. Another similar idea is to incorporate the physical mechanism of traffic flow dynamics as the restriction, such as neural controlled differential equations [180] and Poisson processes [74], to avoid unreasonable predicted values [188] and help to improve the model interpretability.

The last but not the least research opportunity is the combination of graph neural networks and reinforcement learning, which are rarely considered in the surveyed studies, with only one exception [80]. The ensemble of these two models can sometimes create resplendent sparks. For example, some relevant studies leverage reinforcement learning techniques for a more efficient graph neural network structure search [211]. On the other hand, reinforcement learning itself is useful for making optimal decisions in the traffic domain with proper-designed rewards, e.g., traffic light control and autonomous driving. There is still a large research gap in applying reinforcement learning to graph data structures [212,213].

To sum up, the first and second research opportunities are proposed for solving the first research challenge. The third and fourth research opportunities are proposed for solving the second research challenge. The fifth research opportunity is proposed for solving the third research challenge. And the last research opportunity is proposed to inspire more research ideas.

5. Conclusion

A fast-growing number of studies, with the theme of applying graph neural networks for traffic forecasting, are published in 2022. In this survey, we summarized the progression made by these studies and listed their targeted problem, graph type, dataset used and neural networks used. We observed that road traffic flow and speed prediction problems are still the most popular traffic forecasting problems. And the GNN family, GCN and GAT, is one of the promising solution to these problems. To further inspire follow-up studies, new dataset and code resource collections are presented. Research challenges and opportunities are further discussed in this study.

Author Contributions: Conceptualization, W.J., J.L. and M.H.; methodology, W.J., J.L. and M.H.; software, W.J., J.L. and M.H.; validation, W.J., J.L. and M.H.; formal analysis, W.J., J.L. and M.H.; investigation, W.J., J.L. and M.H.; resources,W.J., J.L. and M.H.; data curation, W.J., J.L. and M.H.; writing—original draft preparation, W.J., J.L. and M.H.; writing—review and editing, W.J., J.L. and M.H.; visualization, W.J., J.L. and M.H.; supervision, W.J. and W.G.; project administration, W.J. and W.G.; funding acquisition, W.J. All authors have read and agreed to the published version of the manuscript.

Funding: This research is supported by the Fundamental Research Funds for the Central Universities.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data is available here: https://github.com/jwwthu/GNN4Traffic.

Acknowledgments: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A. The abbreviation list.

The abbreviations used in this manuscript are listed in Table A1 with their full names.

Table A1. The abbreviations used in this manuscript.

Abbreviation	Full Name
AARGNN [87]	Attentive Attributed Recurrent Graph Neural Network

Continued on next page

Table A1 – continued from previous page

Abbreviation	Full Name
ABSTGCN-EF [98]	Attention Based SpatioTemporal Graph Convolutional Network considering External Factors
ADSTGCN [97]	Attention-based Dynamic Spatial–temporal Graph Convolutional Network
ADVW-Net [89]	Adaptive Dual-View WaveNet
AED-DGCN-TSC [93]	Attention Encoder-Decoder Dual Graph Convolutional Network with Time Series Correlation
AGCN-T [92]	Attention based Graph Convolution Network and Transformer
AGCSCN [132]	Adaptive Graph Cross Strided Convolution Network
AM-RGCN [99]	Augmented Multi-component Recurrent Graph Convolutional Network
ASTGAT [96]	Attention-based Spatiotemporal Graph Attention Network
ASTTN [168]	Adaptive Graph Spatial-Temporal Transformer Network
AW-MV-G2S [118]	Attention-weighted Multi-view Graph to Sequence Learning
Ada-STNet [176]	AdaBoost Spatio-Temporal Network
Ada-STNet [91]	Adaptive Spatio-Temporal Graph Neural Network
AdapGL [88]	Adaptive Graph Learning
Bi-GRCN [100]	Bidirectional-graph recurrent convolutional network
CGLGCN [95]	Causal Gated Low-pass Graph Convolution Neural Network
CMGAT [90]	Co-Modal Graph Attention neTwork
CNN	Convolutional Neural Network
CRF	Conditional Random Field
ConvGRU	Convolutional GRU
ConvLSTM	Convolutional LSTM
D^2STGNN [177]	Decoupled Dynamic Spatial-Temporal Graph Neural Network
DCNN [214]	Diffusion Convolutional Neural Network
DDP-GCN [105]	Distance, Direction, and Positional relationship Graph Convolutional Network
DDSTGCN [50]	Dual Dynamic Spatial-Temporal Graph Convolution Network
DG^{2}RNN [71]	Dual Graph Gated Recurrent Neural Network
DGGP [106]	Deep Graph Gaussian Process
DMGC-GAN [75]	Dynamic Multi-Graph Convolutional Network with Generative Adversarial Network
DMVST-VGNN [62]	Deep Multi-View Spatiotemporal Virtual Graph Neural Network
DRL	Deep Reinforcement Learning
DSTAGCN [107]	Dynamic Spatial-Temporal Adjacent Graph Convolutional Network
DSTAGNN [178]	Dynamic Spatial-temporal Aware Graph Neural Network
DSTGCN [108]	Dynamic Spatial-Temporal Graph Convolutional Network
DSTGNN [74]	Dynamical Spatial-Temporal Graph Neural Network
ED2GCN [190]	Two-stage Stacked Graph Convolution Network
ESTNet [112]	Embedded Spatial-temporal Network
ETC	Electronic Toll Collection
FOGS [179]	First-Order Gradient Supervision
Fdsa-STG [115]	Fully Dynamic Self-Attention Spatio-Temporal Graph Network
FedSTN [116]	Federated Deep Learning based on the Spatial-Temporal Long and Short-Term Network
GAMCN [78]	Graph and Attentive Multi-path Convolutional Network
GAN	Generative Adversarial Network
GAT	Graph Attention Network
GCAR [123]	Graph Correlated Attention Recurrent Neural Network

Continued on next page

Table A1 – continued from previous page

Abbreviation	Full Name
GCN	Graph Convolutional Network
GCN-GAN [63]	Graph Convolution and Generative Adversarial Neural Network
GDFormer [121]	Graph Diffusing transFormer
GLU	Gated Linear Unit
GRU	Gated Recurrent Unit
GSeqAtt [124]	Graph sequence neural network with an attention mechanism
GT-GCN [139]	Gated Temporal Graph Convolution Network
GTU	Gated Tanh Unit
HMIAN [64]	Hierarchical Mapping and Interactive Attention Network
IGCRRN [127]	Improved Graph Convolution Res-Recurrent Network
ITS	Intelligent Transportation Systems
IoT	Internet of Things
IoV	Internet of Vehicles
KGR-STGNN [53]	Knowledge Graph Representation Learning and Spatiotemporal Graph Neural Network
kNN	K-Nearest Neighbor
LST-GCN [133]	Long Short-Term Memory Embedded Graph Convolution Network
LSTM	Long Short-term Memory
MA-STN [67]	Multi-dimensional Attention-based Spatial-temporal Network
MADGCN [79]	Multi-Attention Dynamic Graph Convolution Network
MAEGCLSTM [135]	Memory Attention Enhanced Graph Convolution Long Short-term Memory Network
MAGCN [54]	Multi-Attribute Graph Convolutional Network
MAST-GCN [140]	Multigraph Aggregation Spatiotemporal Graph Convolutional Network
MDRGCN [144]	Multi-mode Dynamic Residual Graph Convolution Network
MFDGCN [136]	Multi-Stage Spatio-Temporal Fusion Diffusion Graph Convolutional Network
MG-GAN [120]	Multiple Graph-based Generative Adversarial Network
MHODE [68]	Mixed Hop Diffuse Ordinary Differential Equation
MLP	Multilayer Perceptron
MPNN	Message Passing Neural Network
MS-GAT [130]	Multi-relational Synchronous Graph Attention Network
MSASGCN [137]	Multi-Head Self-Attention Spatiotemporal Graph Convolutional Network
MSDR [182]	Multi-step Dependency Relation Network
MT-HGCN [147]	Multi-task Hypergraph Convolutional Neural Network
MTMGNN [138]	Multi-time Multi-graph Neural Network
MVB-STNet [56]	Multi-View Bayesian Spatio-Temporal Graph Neural Network
MVDSTN [167]	Multi-view Deep Spatio-temporal Network
MVST-GNN [146]	Multi-View Spatial–Temporal Graph Neural Network
PM-MemNet [192]	Pattern-Matching Memory Network
RGSL [186]	Regularized Graph Structure Learning
SAGCN-SST [73]	Self-Attention Graph Convolutional Network with Spatial, Sub-spatial and Temporal blocks
ST-GAT [172]	Spatio-Temporal Graph Attention Network
ST-GCN [125]	Spatial-temporal Graph Convolutional Network
ST-LGSL [187]	Spatio-Temporal Latent Graph Structure Learning
ST-MGAT [159]	Spatio-temporal Multi-head Graph Attention Network
ST-MGCWN [66]	Spatial-temporal Multi-graph Convolutional Wavelet Network

Continued on next page

Table A1 – continued from previous page

Abbreviation	Full Name
ST-MRGNN [129]	Multi-relational Spatiotemporal Graph Neural Network
STAG-GCN [134]	Spatiotemporal Adaptive Gated Graph Convolution Network
STAGCN-EC [152]	Spatial-Temporal Attention Graph Convolution Network on Edge Cloud
STAGCN [51]	Spatio-Temporal Adaptive Graph Convolutional Network
STCAGCN [170]	Spatial-temporal Channel-attention based Graph Convolutional Network
STDEN [188]	Spatio-Temporal Differential Equation Network
STG-NCDE [180]	Spatio-temporal Graph Neural Controlled Differential Equation
STHAN [58]	Spatio-Temporal Heterogeneous Graph Attention Network
STHGCN [59]	Spatiotemporal Prediction Framework using High-order Graph Convolutional Network
STSGAN [158]	Spatial-Temporal Global Semantic Graph Attention Convolution Network
STSSN [86]	Spatio-temporal Sequence-to-sequence Network
STUGCN [153]	Spatial-temporal Upsampling Graph Convolutional Network
STZINBGNN [191]	Spatial-Temporal Zero-Inflated Negative Binomial Graph Neural Network
Seq2Seq	Sequence to Sequence
T-ISTGNN [57]	Transferable Federated Inductive Spatial-Temporal Graph Neural Network
TAMP-S2GCNets [189]	Time-Aware Multipersistence Spatio-Supra Graph Convolutional Network
TCN	Temporal Convolutional Network
TEGCRN [164]	Time-Evolving Graph Convolutional Recurrent Network
TGACN [54]	Temporal Graph Attention Convolutional Neural Network
TGAE [61]	Temporal Graph Autoencoder

Appendix B. The source journal list. 305

The source journals for the surveyed studies are listed in Table A2 with the number of 306
papers counted. 307

Table A2. The source journals for the surveyed papers.

Journal Name	Number of Surveyed Papers
IEEE Transactions on Intelligent Transportation Systems	23
Information Sciences	8
Applied Intelligence	6
Journal of Advanced Transportation	6
Electronics	5
Physica A: Statistical Mechanics and its Applications	5
ACM Transactions on Intelligent Systems and Technology	4
Applied Sciences	4
Knowledge-Based Systems	4
Transportation Research Part C: Emerging Technologies	4
Expert Systems with Applications	3
ACM Transactions on Knowledge Discovery from Data	2
GeoInformatica	2
IEEE Internet of Things Journal	2

Continued on next page

Table A2 – continued from previous page

Journal Name	Number of Surveyed Papers
IEEE Transactions on Knowledge and Data Engineering	2
IET Intelligent Transport Systems	2
ISPRS International Journal of Geo-Information	2
Neural Computing and Applications	2
Wireless Communications and Mobile Computing	2
World Wide Web	2
Applied Soft Computing	1
Big Data	1
Computer Communications	1
Computers, Environment and Urban Systems	1
Connection Science	1
Digital Communications and Networks	1
Digital Signal Processing	1
Engineering Applications of Artificial Intelligence	1
Environment, Development and Sustainability	1
Future Generation Computer Systems	1
IEEE Access	1
IEEE Sensors Journal	1
IEEE Transactions on Big Data	1
IEEE Transactions on Neural Networks and Learning Systems	1
IEEE Transactions on Vehicular Technology	1
International Journal of Intelligent Systems	1
International Journal of Machine Learning and Cybernetics	1
Journal of King Saud University-Computer and Information Sciences	1
Journal of Rail Transport Planning & Management	1
Mathematics	1
Neural Processing Letters	1
Neurocomputing	1
Pattern Recognition Letters	1
Remote Sensing	1
Sustainability	1
Sustainable Computing: Informatics and Systems	1
The Computer Journal	1
Transportation Research Record	1
Transportmetrica B: Transport Dynamics	1
Transportmetrica B: transport dynamics	1

Appendix C. The source conference list. 308

The source conferences for the surveyed papers are listed in Table A3 with the number 309
of papers counted. 310

Table A3. The source conferences for the surveyed papers.

Conference Name	Number of Surveyed Papers
International Joint Conference on Neural Networks (IJCNN)	6
ACM International Conference on Information and Knowledge Management (CIKM)	4

Continued on next page

Table A3 – continued from previous page

Conference Name	Number of Surveyed Papers
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)	4
International Joint Conference on Artificial Intelligence (IJCAI)	2
AAAI Conference on Artificial Intelligence (AAAI)	2
International Conference on Learning Representations (ICLR)	2
IEEE Symposium on Computers and Communications (ISCC)	1
International Conference on Artificial Neural Networks (ICANN)	1
IEEE International Conference on Data Engineering (ICDE)	1
Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)	1
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)	1
International Conference on Very Large Databases (VLDB)	1
International Conference on Machine Learning (ICML)	1
IEEE Wireless Communications and Networking Conference (WCNC)	1
International Conference on Database Systems for Advanced Applications (DASFAA)	1
IEEE International Conference on Computer Supported Cooperative Work in Design (CSCWD)	1

References

1. Zlatanova, S.; Yan, J.; Wang, Y.; Diakité, A.; Isikdag, U.; Sithole, G.; Barton, J. Spaces in spatial science and urban applications—state of the art review. *ISPRS international journal of geo-information* **2020**, *9*, 58.
2. Rehman, A.; Haseeb, K.; Saba, T.; Lloret, J.; Ahmed, Z. Towards resilient and secure cooperative behavior of intelligent transportation system using sensor technologies. *IEEE Sensors Journal* **2022**, *22*, 7352–7360.
3. Wang, L.; Wang, S.; Yuan, Z.; Peng, L. Analyzing potential tourist behavior using PCA and modified affinity propagation clustering based on Baidu index: Taking Beijing city as an example. *Data Science and Management* **2021**, *2*, 12–19.
4. Ahangar, M.N.; Ahmed, Q.Z.; Khan, F.A.; Hafeez, M. A survey of autonomous vehicles: Enabling communication technologies and challenges. *Sensors* **2021**, *21*, 706.
5. Vlahogianni, E.I.; Karlaftis, M.G.; Golias, J.C. Short-term traffic forecasting: Where we are and where we're going. *Transportation Research Part C: Emerging Technologies* **2014**, *43*, 3–19.
6. Vlahogianni, E.I.; Golias, J.C.; Karlaftis, M.G. Short-term traffic forecasting: Overview of objectives and methods. *Transport reviews* **2004**, *24*, 533–557.
7. Ermagun, A.; Levinson, D. Spatiotemporal traffic forecasting: review and proposed directions. *Transport Reviews* **2018**, *38*, 786–814.
8. Yu, R.; Li, Y.; Shahabi, C.; Demiryurek, U.; Liu, Y. Deep learning: A generic approach for extreme condition traffic forecasting. Proceedings of the 2017 SIAM international Conference on Data Mining. SIAM, 2017, pp. 777–785.
9. He, M.; Gu, W.; Kong, Y.; Zhang, L.; Spanos, C.J.; Mosalam, K.M. Causalbg: Causal recurrent neural network for the blood glucose inference with IoT platform. *IEEE Internet of Things Journal* **2019**, *7*, 598–610.
10. Lana, I.; Del Ser, J.; Velez, M.; Vlahogianni, E.I. Road traffic forecasting: Recent advances and new challenges. *IEEE Intelligent Transportation Systems Magazine* **2018**, *10*, 93–109.
11. Ahmed, M.S.; Cook, A.R. *Analysis of freeway traffic time-series data by using Box-Jenkins techniques*; Number 722, 1979.
12. Levin, M.; Tsao, Y.D. On forecasting freeway occupancies and volumes (abridgment). *Transportation Research Record* **1980**.
13. Moorthy, C.; Ratcliffe, B. Short term traffic forecasting using time series methods. *Transportation planning and technology* **1988**, *12*, 45–56.
14. Nihan, N.L.; Holmesland, K.O. Use of the Box and Jenkins time series technique in traffic forecasting. *Transportation* **1980**, *9*, 125–143.
15. Zhang, J.; Zheng, Y.; Qi, D. Deep spatio-temporal residual networks for citywide crowd flows prediction. Thirty-first AAAI conference on artificial intelligence, 2017.
16. Jiang, W. TaxiBJ21: An open crowd flow dataset based on Beijing taxi GPS trajectories. *Internet Technology Letters* **2022**, *5*, e297.
17. Guo, S.; Lin, Y.; Feng, N.; Song, C.; Wan, H. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. Proceedings of the AAAI conference on artificial intelligence, 2019, Vol. 33, pp. 922–929.
18. Sun, B.; Zhao, D.; Shi, X.; He, Y. Modeling global spatial–temporal graph attention network for traffic prediction. *IEEE Access* **2021**, *9*, 8581–8594.

19. Ang, K.L.M.; Seng, J.K.P.; Ngharamike, E.; Ijemaru, G.K. Emerging Technologies for Smart Cities' Transportation: Geo-Information, Data Analytics and Machine Learning Approaches. *ISPRS International Journal of Geo-Information* **2022**, *11*, 85.
20. Jiang, W. Vehicle Destination Prediction with Spatial Clustering and Machine Learning. *Internet Technology Letters* **2022**, p. e403.
21. Jiang, W.; Zhang, L. Geospatial data to images: A deep-learning framework for traffic forecasting. *Tsinghua Science and Technology* **2018**, *24*, 52–64.
22. Khan, Z.; Khan, S.M.; Dey, K.; Chowdhury, M. Development and evaluation of recurrent neural network-based models for hourly traffic volume and annual average daily traffic prediction. *Transportation Research Record* **2019**, *2673*, 489–503.
23. Jiang, W. Applications of deep learning in stock market prediction: recent progress. *Expert Systems with Applications* **2021**, *184*, 115537.
24. Santhosh, M.; Venkaiah, C.; Vinod Kumar, D. Current advances and approaches in wind speed and wind power forecasting for improved renewable energy integration: A review. *Engineering Reports* **2020**, *2*, e12178.
25. Rajbhandari, Y.; Marahatta, A.; Ghimire, B.; Shrestha, A.; Gachhadar, A.; Thapa, A.; Chapagain, K.; Korba, P. Impact study of temperature on the time series electricity demand of urban nepal for short-term load forecasting. *Applied System Innovation* **2021**, *4*, 43.
26. Shankarnarayan, V.K.; Ramakrishna, H. Comparative study of three stochastic future weather forecast approaches: A case study. *Data Science and Management* **2021**, *3*, 3–12.
27. Zhao, E.; Sun, S.; Wang, S. New developments in wind energy forecasting with artificial intelligence and big data: A scientometric insight. *Data Science and Management* **2022**.
28. Jiang, W. Internet traffic prediction with deep neural networks. *Internet Technology Letters* **2022**, *5*, e314.
29. Jiang, W. Internet traffic matrix prediction with convolutional LSTM neural network. *Internet Technology Letters* **2022**, *5*, e322.
30. Sousa, M.; Tomé, A.M.; Moreira, J. Long-term forecasting of hourly retail customer flow on intermittent time series with multiple seasonality. *Data Science and Management* **2022**, *5*, 137–148.
31. Jiang, W. Deep learning based short-term load forecasting incorporating calendar and weather information. *Internet Technology Letters* **2022**, *5*, e383.
32. Zhuang, X.; Yu, Y.; Chen, A. A combined forecasting method for intermittent demand using the automotive aftermarket data. *Data Science and Management* **2022**.
33. Jiang, W. Cellular traffic prediction with machine learning: A survey. *Expert Systems with Applications* **2022**, p. 117163.
34. Tedjopurnomo, D.A.; Bao, Z.; Zheng, B.; Choudhury, F.; Qin, A.K. A survey on modern deep neural network for traffic prediction: Trends, methods and challenges. *IEEE Transactions on Knowledge and Data Engineering* **2020**.
35. Boukerche, A.; Wang, J. Machine Learning-based traffic prediction models for Intelligent Transportation Systems. *Computer Networks* **2020**, *181*, 107530.
36. Boukerche, A.; Tao, Y.; Sun, P. Artificial intelligence-based vehicular traffic flow prediction methods for supporting intelligent transportation systems. *Computer networks* **2020**, *182*, 107484.
37. Manibardo, E.L.; Laña, I.; Del Ser, J. Deep learning for road traffic forecasting: Does it make a difference? *IEEE Transactions on Intelligent Transportation Systems* **2021**.
38. Lee, K.; Eo, M.; Jung, E.; Yoon, Y.; Rhee, W. Short-term traffic prediction with deep neural networks: A survey. *IEEE Access* **2021**, *9*, 54739–54756.
39. Yin, X.; Wu, G.; Wei, J.; Shen, Y.; Qi, H.; Yin, B. Deep learning on traffic prediction: Methods, analysis and future directions. *IEEE Transactions on Intelligent Transportation Systems* **2021**.
40. Jiang, W.; Luo, J. Big data for traffic estimation and prediction: a survey of data and tools. *Applied System Innovation* **2022**, *5*, 23.
41. Jiang, W. Bike sharing usage prediction with deep learning: a survey. *Neural Computing and Applications* **2022**, pp. 1–17.
42. Ye, J.; Zhao, J.; Ye, K.; Xu, C. How to build a graph-based deep learning architecture in traffic domain: A survey. *IEEE Transactions on Intelligent Transportation Systems* **2022**, *23*, 3904–3924.
43. Jiang, W.; Luo, J. Graph neural network for traffic forecasting: A survey. *Expert Systems with Applications* **2022**, p. 117921.
44. Kipf, T.N.; Welling, M. Semi-supervised classification with graph convolutional networks. International Conference on Learning Representations (ICLR '17), 2017.
45. Veličković, P.; Cucurull, G.; Casanova, A.; Romero, A.; Liò, P.; Bengio, Y. Graph Attention Networks. International Conference on Learning Representations, 2018.
46. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. *Advances in neural information processing systems* **2017**, *30*.
47. Wu, Z.; Pan, S.; Chen, F.; Long, G.; Zhang, C.; Philip, S.Y. A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems* **2020**, *32*, 4–24.
48. Zhang, Z.; Cui, P.; Zhu, W. Deep learning on graphs: A survey. *IEEE Transactions on Knowledge and Data Engineering* **2020**.
49. Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; Zhang, W. Informer: Beyond efficient transformer for long sequence time-series forecasting. Proceedings of the AAAI Conference on Artificial Intelligence, 2021, Vol. 35, pp. 11106–11115.
50. Sun, Y.; Jiang, X.; Hu, Y.; Duan, F.; Guo, K.; Wang, B.; Gao, J.; Yin, B. Dual Dynamic Spatial-Temporal Graph Convolution Network for Traffic Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
51. Ma, Q.; Sun, W.; Gao, J.; Ma, P.; Shi, M. Spatio-temporal adaptive graph convolutional networks for traffic flow forecasting. *IET Intelligent Transport Systems* **2022**.

52. Kong, X.; Wei, X.; Zhang, J.; Xing, W.; Lu, W. JointGraph: joint pre-training framework for traffic forecasting with spatial-temporal gating diffusion graph attention network. *Applied Intelligence* **2022**, pp. 1–18.
53. Wang, S.; Lv, Y.; Peng, Y.; Piao, X.; Zhang, Y. Metro Traffic Flow Prediction via Knowledge Graph and Spatiotemporal Graph Neural Network. *Journal of Advanced Transportation* **2022**.
54. Wang, Y.; Zhao, A.; Li, J.; Lv, Z.; Dong, C.; Li, H. Multi-attribute Graph Convolution Network for Regional Traffic Flow Prediction. *Neural Processing Letters* **2022**, pp. 1–27.
55. Zhang, L.; Geng, X.; Qin, Z.; Wang, H.; Wang, X.; Zhang, Y.; Liang, J.; Wu, G.; Song, X.; Wang, Y. Multi-modal graph interaction for multi-graph convolution network in urban spatiotemporal forecasting. *Sustainability* **2022**, *14*, 12397.
56. Xia, J.; Wang, S.; Wang, X.; Xia, M.; Xie, K.; Cao, J. Multi-view Bayesian spatio-temporal graph neural networks for reliable traffic flow prediction. *International Journal of Machine Learning and Cybernetics* **2022**, pp. 1–14.
57. Qi, Y.; Wu, J.; Bashir, A.K.; Lin, X.; Yang, W.; Alshehri, M.D. Privacy-Preserving Cross-Area Traffic Forecasting in ITS: A Transferable Spatial-Temporal Graph Neural Network Approach. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
58. Ling, S.; Yu, Z.; Cao, S.; Zhang, H.; Hu, S. STHAN: Transportation Demand Forecasting with Compound Spatio-Temporal Relationships. *ACM Transactions on Knowledge Discovery from Data (TKDD)* **2022**.
59. Wang, J.; Wang, W.; Yu, W.; Liu, X.; Jia, K.; Li, X.; Zhong, M.; Sun, Y.; Xu, Y. STHGCN: A spatiotemporal prediction framework based on higher-order graph convolution networks. *Knowledge-Based Systems* **2022**, p. 109985.
60. Dai, S.; Wang, J.; Huang, C.; Yu, Y.; Dong, J. Dynamic Multi-View Graph Neural Networks for Citywide Traffic Inference. *ACM Transactions on Knowledge Discovery from Data (TKDD)* **2022**.
61. Wang, Q.; Jiang, H.; Qiu, M.; Liu, Y.; Ye, D. TGAE: Temporal Graph Autoencoder for Travel Forecasting. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
62. Jin, G.; Xi, Z.; Sha, H.; Feng, Y.; Huang, J. Deep multi-view graph-based network for citywide ride-hailing demand prediction. *Neurocomputing* **2022**, *510*, 79–94.
63. Zheng, H.; Li, X.; Li, Y.; Yan, Z.; Li, T. GCN-GAN: Integrating Graph Convolutional Network and Generative Adversarial Network for Traffic Flow Prediction. *IEEE Access* **2022**, *10*, 94051–94062.
64. Sun, J.; Peng, M.; Jiang, H.; Hong, Q.; Sun, Y. HMIAN: a Hierarchical Mapping and Interactive Attention Data Fusion Network for Traffic Forecasting. *IEEE Internet of Things Journal* **2022**.
65. Djenouri, Y.; Belhadi, A.; Srivastava, G.; Lin, J.C.W. Hybrid graph convolution neural network and branch-and-bound optimization for traffic flow forecasting. *Future Generation Computer Systems* **2023**, *139*, 100–108.
66. Xiu, C.; Sun, Y.; Peng, Q. Modelling traffic as multi-graph signals: Using domain knowledge to enhance the network-level passenger flow prediction in metro systems. *Journal of Rail Transport Planning & Management* **2022**, *24*, 100342.
67. Xu, G.; Hu, X. Multi-Dimensional Attention Based Spatial-Temporal Networks for Traffic Forecasting. *Wireless Communications and Mobile Computing* **2022**, *2022*.
68. Huang, X.; Lan, Y.; Ye, Y.; Wang, J.; Jiang, Y. Traffic Flow Prediction Based on Multi-Mode Spatial-Temporal Convolution of Mixed Hop Diffuse ODE. *Electronics* **2022**, *11*, 3012.
69. Ge, Y.; Zhai, J.F.; Su, P.C. Traffic Flow Prediction Based on Multi-Spatiotemporal Attention Gated Graph Convolution Network. *Journal of Advanced Transportation* **2022**.
70. Pan, X.; Hou, F.; Li, S. Traffic Speed Prediction Based on Time Classification in Combination With Spatial Graph Convolutional Network. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
71. Zhao, J.; Chen, C.; Liao, C.; Huang, H.; Ma, J.; Pu, H.; Luo, J.; Zhu, T.; Wang, S. 2F-TP: Learning Flexible Spatiotemporal Dependency for Flexible Traffic Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
72. Qi, X.; Mei, G.; Tu, J.; Xi, N.; Piccialli, F. A Deep Learning Approach for Long-Term Traffic Flow Prediction With Multifactor Fusion Using Spatiotemporal Graph Convolutional Network. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
73. Zheng, G.; Chai, W.K.; Katos, V. A dynamic spatial–temporal deep learning framework for traffic speed prediction on large-scale road networks. *Expert Systems with Applications* **2022**, *195*, 116585.
74. Huang, F.; Yi, P.; Wang, J.; Li, M.; Peng, J.; Xiong, X. A dynamical spatial-temporal graph neural network for traffic demand prediction. *Information Sciences* **2022**, *594*, 286–304.
75. Huang, Z.; Zhang, W.; Wang, D.; Yin, Y. A GAN framework-based dynamic multi-graph convolutional network for origin–destination-based ride-hailing demand prediction. *Information Sciences* **2022**, *601*, 129–146.
76. Jin, J.; Rong, D.; Zhang, T.; Ji, Q.; Guo, H.; Lv, Y.; Ma, X.; Wang, F.Y. A GAN-Based Short-Term Link Traffic Prediction Approach for Urban Road Networks Under a Parallel Learning Framework. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
77. Xu, D.; Lin, Z.; Zhou, L.; Li, H.; Niu, B. A GATs-GAN framework for road traffic states forecasting. *Transportmetrica B: transport dynamics* **2022**, *10*, 718–730.
78. Qi, J.; Zhao, Z.; Tanin, E.; Cui, T.; Nassir, N.; Sarvi, M. A Graph and Attentive Multi-Path Convolutional Network for Traffic Prediction. *IEEE Transactions on Knowledge and Data Engineering* **2022**.
79. Wu, M.; Jia, H.; Luo, D.; Luo, H.; Zhao, F.; Li, G. A multi-attention dynamic graph convolution network with cost-sensitive learning approach to road-level and minute-level traffic accident prediction. *IET Intelligent Transport Systems* **2022**.
80. Shang, P.; Liu, X.; Yu, C.; Yan, G.; Xiang, Q.; Mi, X. A new ensemble deep graph reinforcement learning network for spatio-temporal traffic volume forecasting in a freeway network. *Digital Signal Processing* **2022**, *123*, 103419.

81. Hu, Z.; Shao, F.; Sun, R. A New Perspective on Traffic Flow Prediction: A Graph Spatial-Temporal Network with Complex Network Information. *Electronics* **2022**, *11*, 2432.
82. Chen, Y.; Chen, X.M. A novel reinforced dynamic graph convolutional network model with data imputation for network-wide traffic flow prediction. *Transportation Research Part C: Emerging Technologies* **2022**, *143*, 103820.
83. Diao, C.; Zhang, D.; Liang, W.; Li, K.C.; Hong, Y.; Gaudiot, J.L. A Novel Spatial-Temporal Multi-Scale Alignment Graph Neural Network Security Model for Vehicles Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
84. Liu, F.; Wang, J.; Tian, J.; Zhuang, D.; Miranda-Moreno, L.; Sun, L. A Universal Framework of Spatiotemporal Bias Block for Long-Term Traffic Forecasting. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
85. Li, Y.; Zhao, W.; Fan, H. A Spatio-Temporal Graph Neural Network Approach for Traffic Flow Prediction. *Mathematics* **2022**, *10*, 1754.
86. Cao, S.; Wu, L.; Wu, J.; Wu, D.; Li, Q. A spatio-temporal sequence-to-sequence network for traffic flow prediction. *Information Sciences* **2022**, *610*, 185–203.
87. Chen, L.; Shao, W.; Lv, M.; Chen, W.; Zhang, Y.; Yang, C. AARGNN: An Attentive Attributed Recurrent Graph Neural Network for Traffic Flow Prediction Considering Multiple Dynamic Factors. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
88. Zhang, W.; Zhu, F.; Lv, Y.; Tan, C.; Liu, W.; Zhang, X.; Wang, F.Y. AdapGL: An adaptive graph learning algorithm for traffic prediction based on spatiotemporal neural networks. *Transportation Research Part C: Emerging Technologies* **2022**, *139*, 103659.
89. Jin, G.; Liu, C.; Xi, Z.; Sha, H.; Liu, Y.; Huang, J. Adaptive Dual-View WaveNet for urban spatial–temporal event prediction. *Information Sciences* **2022**, *588*, 315–330.
90. Xu, H.; Zou, T.; Liu, M.; Qiao, Y.; Wang, J.; Li, X. Adaptive Spatiotemporal Dependence Learning for Multi-Mode Transportation Demand Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
91. Ta, X.; Liu, Z.; Hu, X.; Yu, L.; Sun, L.; Du, B. Adaptive Spatio-temporal Graph Neural Network for traffic forecasting. *Knowledge-Based Systems* **2022**, *242*, 108199.
92. Feng, J.; Yu, L.; Ma, R. AGCN-T: A Traffic Flow Prediction Model for Spatial-Temporal Network Dynamics. *Journal of Advanced Transportation* **2022**, *2022*.
93. Zhao, S.; Li, X. An Attention Encoder-Decoder Dual Graph Convolutional Network with Time Series Correlation for Multi-Step Traffic Flow Prediction. *Journal of Advanced Transportation* **2022**, *2022*.
94. Liao, L.; Hu, Z.; Zheng, Y.; Bi, S.; Zou, F.; Qiu, H.; Zhang, M. An improved dynamic Chebyshev graph convolution network for traffic flow prediction with spatial-temporal attention. *Applied Intelligence* **2022**, pp. 1–13.
95. Xu, X.; Mao, H.; Zhao, Y.; Lü, X. An Urban Traffic Flow Fusion Network Based on a Causal Spatiotemporal Graph Convolution Network. *Applied Sciences* **2022**, *12*, 7010.
96. Wang, Y.; Jing, C.; Xu, S.; Guo, T. Attention based spatiotemporal graph attention networks for traffic flow forecasting. *Information Sciences* **2022**, *607*, 869–883.
97. Zhao, J.; Liu, Z.; Sun, Q.; Li, Q.; Jia, X.; Zhang, R. Attention-based dynamic spatial-temporal graph convolutional networks for traffic speed forecasting. *Expert Systems with Applications* **2022**, *204*, 117511.
98. Ye, J.; Xue, S.; Jiang, A. Attention-based spatio-temporal graph convolutional network considering external factors for multi-step traffic flow prediction. *Digital Communications and Networks* **2022**, *8*, 343–350.
99. Zhang, C.; Zhou, H.Y.; Qiu, Q.; Jian, Z.; Zhu, D.; Cheng, C.; He, L.; Liu, G.; Wen, X.; Hu, R. Augmented Multi-Component Recurrent Graph Convolutional Network for Traffic Flow Forecasting. *ISPRS International Journal of Geo-Information* **2022**, *11*, 88.
100. Jiang, W.; Xiao, Y.; Liu, Y.; Liu, Q.; Li, Z. Bi-GRCN: A Spatio-Temporal Traffic Flow Prediction Model Based on Graph Neural Network. *Journal of Advanced Transportation* **2022**, *2022*.
101. Hu, C.; Ning, B.; Gu, Q.; Qu, J.; Jeon, S.; Du, B. Big data analytics-based traffic flow forecasting using inductive spatial-temporal network. *Environment, Development and Sustainability* **2022**, pp. 1–17.
102. Yin, G.; Huang, Z.; Bao, Y.; Wang, H.; Li, L.; Ma, X.; Zhang, Y. ConvGCN-RF: A hybrid learning model for commuting flow prediction considering geographical semantics and neighborhood effects. *GeoInformatica* **2022**, pp. 1–21.
103. Zhao, T.; Huang, Z.; Tu, W.; He, B.; Cao, R.; Cao, J.; Li, M. Coupling graph deep learning and spatial-temporal influence of built environment for short-term bus travel demand prediction. *Computers, Environment and Urban Systems* **2022**, *94*, 101776.
104. Li, F.; Feng, J.; Yan, H.; Jin, D.; Li, Y. Crowd Flow Prediction for irregular Regions with Semantic Graph Attention Network. *ACM Transactions on Intelligent Systems and Technology (TIST)* **2022**.
105. Lee, K.; Rhee, W. DDP-GCN: Multi-graph convolutional network for spatiotemporal traffic forecasting. *Transportation Research Part C: Emerging Technologies* **2022**, *134*, 103466.
106. Jiang, Y.; Fan, J.; Liu, Y.; Zhang, X. Deep Graph Gaussian Processes for Short-Term Traffic Flow Forecasting From Spatiotemporal Data. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
107. Zheng, Q.; Zhang, Y. DSTAGCN: Dynamic Spatial-Temporal Adjacent Graph Convolutional Network for Traffic Forecasting. *IEEE Transactions on Big Data* **2022**.
108. Hu, J.; Lin, X.; Wang, C. DSTGCN: Dynamic Spatial-Temporal Graph Convolutional Network for Traffic Prediction. *IEEE Sensors Journal* **2022**, *22*, 13116–13124.
109. Zhang, W.; Zhu, K.; Zhang, S.; Chen, Q.; Xu, J. Dynamic graph convolutional networks based on spatiotemporal data embedding for traffic flow forecasting. *Knowledge-Based Systems* **2022**, p. 109028.

110. Liu, Z.; Bian, J.; Zhang, D.; Chen, Y.; Shen, G.; Kong, X. Dynamic Multi-View Coupled Graph Convolution Network for Urban Travel Demand Forecasting. *Electronics* **2022**, *11*, 2620.
111. Han, S.Y.; Zhao, Q.; Sun, Q.W.; Zhou, J.; Chen, Y.H. EnGS-DGR: Traffic Flow Forecasting with Indefinite Forecasting Interval by Ensemble GCN, Seq2Seq, and Dynamic Graph Reconfiguration. *Applied Sciences* **2022**, *12*, 2890.
112. Luo, G.; Zhang, H.; Yuan, Q.; Li, J.; Wang, F.Y. ESTNet: Embedded Spatial-Temporal Network for Modeling Traffic Flow Dynamics. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
113. Kong, X.; Wang, K.; Hou, M.; Xia, F.; Karmakar, G.; Li, J. Exploring Human Mobility for Multi-Pattern Passenger Prediction: A Graph Learning Framework. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
114. Zou, F.; Ren, Q.; Tian, J.; Guo, F.; Huang, S.; Liao, L.; Wu, J. Expressway Speed Prediction Based on Electronic Toll Collection Data. *Electronics* **2022**, *11*, 1613.
115. Duan, Y.; Chen, N.; Shen, S.; Zhang, P.; Qu, Y.; Yu, S. Fdsa-STG: Fully Dynamic Self-Attention Spatio-Temporal Graph Networks for Intelligent Traffic Flow Prediction. *IEEE Transactions on Vehicular Technology* **2022**.
116. Yuan, X.; Chen, J.; Yang, J.; Zhang, N.; Yang, T.; Han, T.; Taherkordi, A. FedSTN: Graph Representation Driven Federated Learning for Edge Computing Enabled Urban Traffic Flow Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
117. Jia, T.; Cai, C. Forecasting citywide short-term turning traffic flow at intersections using an attention-based spatiotemporal deep learning model. *Transportmetrica B: Transport Dynamics* **2022**, pp. 1–23.
118. Bao, J.; Kang, J.; Yang, Z.; Chen, X. Forecasting network-wide multi-step metro ridership with an attention-weighted multi-view graph to sequence learning approach. *Expert Systems with Applications* **2022**, *210*, 118475.
119. Zhang, X.; Xu, Y.; Shao, Y. Forecasting traffic flow with spatial–temporal convolutional graph attention networks. *Neural Computing and Applications* **2022**, pp. 1–23.
120. Liu, C.; Xiao, Z.; Wang, D.; Cheng, M.; Chen, H.; Cai, J. Foreseeing private car transfer between urban regions with multiple graph-based generative adversarial networks. *World Wide Web* **2022**, pp. 1–20.
121. Su, J.; Jin, Z.; Ren, J.; Yang, J.; Liu, Y. GDFormer: A Graph Diffusing Attention based approach for Traffic Flow Prediction. *Pattern Recognition Letters* **2022**, *156*, 126–132.
122. James, J. Graph Construction for Traffic Prediction: A Data-Driven Approach. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
123. Geng, X.; He, X.; Xu, L.; Yu, J. Graph correlated attention recurrent neural network for multivariate time series forecasting. *Information Sciences* **2022**, *606*, 126–142.
124. Lu, Z.; Lv, W.; Xie, Z.; Du, B.; Xiong, G.; Sun, L.; Wang, H. Graph Sequence Neural Network with an Attention Mechanism for Traffic Speed Prediction. *ACM Transactions on Intelligent Systems and Technology (TIST)* **2022**, *13*, 1–24.
125. Wang, H.; Zhang, R.; Cheng, X.; Yang, L. Hierarchical traffic flow prediction based on spatial-temporal graph convolutional network. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
126. Dai, F.; Cao, P.; Huang, P.; Mo, Q.; Huang, B. Hybrid deep learning approach for traffic speed prediction. *Big Data* **2022**.
127. Zhang, Q.; Yin, C.; Chen, Y.; Su, F. IGCRRN: Improved Graph Convolution Res-Recurrent Network for spatio-temporal dependence capturing and traffic flow prediction. *Engineering Applications of Artificial Intelligence* **2022**, *114*, 105179.
128. Jiang, M.; Li, C.; Li, K.; Yang, Z.; Liu, H. Inter-Block Flow Prediction with Relation Graph Network for Cold-start on Bike-Sharing System. *IEEE Internet of Things Journal* **2022**.
129. Liang, Y.; Huang, G.; Zhao, Z. Joint demand prediction for multimodal systems: A multi-task multi-relational spatiotemporal graph neural network approach. *Transportation Research Part C: Emerging Technologies* **2022**, *140*, 103731.
130. Huang, J.; Luo, K.; Cao, L.; Wen, Y.; Zhong, S. Learning Multiaspect Traffic Couplings by Multirelational Graph Attention Networks for Traffic Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
131. Xu, M.; Li, X.; Wang, F.; Shang, J.S.; Chong, T.; Cheng, W.; Xu, J. Learning to effectively model spatial-temporal heterogeneity for traffic flow forecasting. *World Wide Web* **2022**, pp. 1–17.
132. Li, Z.; Zhang, Y.; Guo, D.; Zhou, X.; Wang, X.; Zhu, L. Long-term traffic forecasting based on adaptive graph cross strided convolution network. *Applied Intelligence* **2022**, pp. 1–15.
133. Han, X.; Gong, S. LST-GCN: Long Short-Term Memory Embedded Graph Convolution Network for Traffic Flow Forecasting. *Electronics* **2022**, *11*, 2230.
134. Lu, B.; Gan, X.; Jin, H.; Fu, L.; Wang, X.; Zhang, H. Make More Connections: Urban Traffic Flow Forecasting with Spatiotemporal Adaptive Gated Graph Convolution Network. *ACM Transactions on Intelligent Systems and Technology (TIST)* **2022**, *13*, 1–25.
135. Qin, Y.; Zhao, F.; Fang, Y.; Luo, H.; Wang, C. Memory attention enhanced graph convolution long short-term memory network for traffic forecasting. *International Journal of Intelligent Systems* **2022**, *37*, 6555–6576.
136. Cui, Z.; Zhang, J.; Noh, G.; Park, H.J. MFDGCN: Multi-Stage Spatio-Temporal Fusion Diffusion Graph Convolutional Network for Traffic Prediction. *Applied Sciences* **2022**, *12*, 2688.
137. Cao, Y.; Liu, D.; Yin, Q.; Xue, F.; Tang, H. MSASGCN: Multi-Head Self-Attention Spatiotemporal Graph Convolutional Network for Traffic Flow Forecasting. *Journal of Advanced Transportation* **2022**, *2022*.
138. Yin, D.; Jiang, R.; Deng, J.; Li, Y.; Xie, Y.; Wang, Z.; Zhou, Y.; Song, X.; Shang, J.S. MTMGNN: Multi-time multi-graph neural network for metro passenger flow prediction. *GeoInformatica* **2022**, pp. 1–29.
139. Feng, H.; Jiang, X. Multi-step ahead traffic speed prediction based on gated temporal graph convolution network. *Physica A: Statistical Mechanics and its Applications* **2022**, *606*, 128075.

140. Li, C.; Zhang, H.; Wang, Z.; Wu, Y.; Yang, F. Multigraph Aggregation Spatiotemporal Graph Convolution Network for Ride-Hailing Pick-Up Region Prediction. *Wireless Communications and Mobile Computing* **2022**, *2022*.
141. Wang, S.; Zhang, M.; Miao, H.; Peng, Z.; Yu, P.S. Multivariate Correlation-aware Spatio-temporal Graph Convolutional Networks for Multi-scale Traffic Prediction. *ACM Transactions on Intelligent Systems and Technology (TIST)* **2022**, *13*, 1–22.
142. Zhao, C.; Li, X.; Shao, Z.; Yang, H.; Wang, F. Multi-featured spatial-temporal and dynamic multi-graph convolutional network for metro passenger flow prediction. *Connection Science* **2022**, *34*, 1252–1272.
143. Sun, Y.; Jiang, G.; Lam, S.K.; He, P.; Ning, F. Multi-fold Correlation Attention Network for Predicting Traffic Speeds with Heterogeneous Frequency. *Applied Soft Computing* **2022**, p. 108977.
144. Huang, X.; Ye, Y.; Ding, W.; Yang, X.; Xiong, L. Multi-mode dynamic residual graph convolution network for traffic flow prediction. *Information Sciences* **2022**, *609*, 548–564.
145. Wang, Y.; Qin, Y.; Guo, J.; Cao, Z.; Jia, L. Multi-point short-term prediction of station passenger flow based on temporal multi-graph convolutional network. *Physica A: Statistical Mechanics and its Applications* **2022**, *604*, 127959.
146. Li, H.; Jin, D.; Li, X.; Huang, H.; Yun, J.; Huang, L. Multi-View Spatial–Temporal Graph Neural Network for Traffic Prediction. *The Computer Journal* **2022**.
147. Wang, J.; Zhang, Y.; Wang, L.; Hu, Y.; Piao, X.; Yin, B. Multitask Hypergraph Convolutional Networks: A Heterogeneous Traffic Prediction Framework. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
148. Yang, H.; Zhang, X.; Li, Z.; Cui, J. Region-Level Traffic Prediction Based on Temporal Multi-Spatial Dependence Graph Convolutional Network from GPS Data. *Remote Sensing* **2022**, *14*, 303.
149. Abdelraouf, A.; Abdel-Aty, M.; Mahmoud, N. Sequence-to-Sequence Recurrent Graph Convolutional Networks for Traffic Estimation and Prediction Using Connected Probe Vehicle Data. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
150. Baghbani, A.; Bouguila, N.; Patterson, Z. Short-Term Passenger Flow Prediction Using a Bus Network Graph Convolutional Long Short-Term Memory Neural Network Model. *Transportation Research Record* **2022**, p. 03611981221112673.
151. Xia, M.; Jin, D.; Chen, J. Short-Term Traffic Flow Prediction Based on Graph Convolutional Networks and Federated Learning. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
152. Lai, Q.; Tian, J.; Wang, W.; Hu, X. Spatial-Temporal Attention Graph Convolution Network on Edge Cloud for Traffic Flow Prediction. *IEEE Transactions on Intelligent Transportation Systems* **2022**.
153. Zhang, R.; Xie, F.; Sun, R.; Huang, L.; Liu, X.; Shi, J. Spatial-temporal dynamic semantic graph neural network. *Neural Computing and Applications* **2022**, pp. 1–14.
154. Zhang, S.; Liu, Y.; Xiao, Y.; He, R. Spatial-temporal upsampling graph convolutional network for daily long-term traffic speed prediction. *Journal of King Saud University-Computer and Information Sciences* **2022**.
155. Dong, C.; Zhang, K.; Wei, X.; Wang, Y.; Yang, Y. Spatiotemporal Graph Attention Network modeling for multi-step passenger demand prediction at multi-zone level. *Physica A: Statistical Mechanics and its Applications* **2022**, *603*, 127789.
156. Ni, Q.; Zhang, M. STGMN: A gated multi-graph convolutional network framework for traffic flow prediction. *Applied Intelligence* **2022**, pp. 1–14.
157. Ou, J.; Sun, J.; Zhu, Y.; Jin, H.; Liu, Y.; Zhang, F.; Huang, J.; Wang, X. STP-TrellisNets+: Spatial-Temporal Parallel TrellisNets for Multi-Step Metro Station Passenger Flow Prediction. *IEEE Transactions on Knowledge and Data Engineering* **2022**.
158. Zhou, J.; Qin, X.; Yu, K.; Jia, Z.; Du, Y. STSGAN: Spatial-Temporal Global Semantic Graph Attention Convolution Networks for Urban Flow Prediction. *ISPRS International Journal of Geo-Information* **2022**, *11*, 381.
159. Wang, B.; Wang, J. ST-MGAT: Spatio-temporal multi-head graph attention network for Traffic prediction. *Physica A: Statistical Mechanics and its Applications* **2022**, *603*, 127762.
160. Liao, W.; Zeng, B.; Liu, J.; Wei, P.; Cheng, X. Taxi demand forecasting based on the temporal multimodal information fusion graph neural network. *Applied Intelligence* **2022**, pp. 1–14.
161. Zhang, W.; Yan, S.; Li, J. TCP-BAST: A novel approach to traffic congestion prediction with bilateral alternation on spatiality and temporality. *Information Sciences* **2022**, *608*, 718–733.
162. Khaled, A.; Elsir, A.M.T.; Shen, Y. TFGAN: Traffic forecasting using generative adversarial network with multi-graph convolutional network. *Knowledge-Based Systems* **2022**, p. 108990.
163. Chen, L.; Shi, P.; Li, G.; Qi, T. Traffic flow prediction using multi-view graph convolution and masked attention mechanism. *Computer Communications* **2022**, *194*, 446–457.
164. Mai, W.; Chen, J.; Chen, X. Time-Evolving Graph Convolutional Recurrent Network for Traffic Prediction. *Applied Sciences* **2022**, *12*, 2842.
165. Wang, J.; Wang, W.; Liu, X.; Yu, W.; Li, X.; Sun, P. Traffic prediction based on auto spatiotemporal multi-graph adversarial neural network. *Physica A: Statistical Mechanics and its Applications* **2022**, *590*, 126736.
166. Wang, T.; Ni, S.; Qin, T.; Cao, D. TransGAT: A dynamic graph attention residual networks for traffic flow forecasting. *Sustainable Computing: Informatics and Systems* **2022**, *36*, 100779.
167. Wu, Y.; Zhang, H.; Li, C.; Tao, S.; Yang, F. Urban ride-hailing demand prediction with multi-view information fusion deep learning framework. *Applied Intelligence* **2022**, pp. 1–19.
168. Feng, A.; Tassiulas, L. Adaptive Graph Spatial-Temporal Transformer Network for Traffic Forecasting. Proceedings of the 31st ACM International Conference on Information & Knowledge Management, 2022, pp. 3933–3937.

169. Li, F.; Yan, H.; Jin, G.; Liu, Y.; Li, Y.; Jin, D. Automated Spatio-Temporal Synchronous Modeling with Multiple Graphs for Traffic Prediction. Proceedings of the 31st ACM International Conference on Information & Knowledge Management, 2022, pp. 1084–1093.
170. Wang, Y.; Ren, Q. Dynamic Graph Convolutional Network for Long Short-term Traffic Flow Prediction. 2022 IEEE Symposium on Computers and Communications (ISCC). IEEE, 2022, pp. 1–6.
171. Liu, Z.; Fu, K.; Liu, X. Multi-view Cascading Spatial-Temporal Graph Neural Network for Traffic Flow Forecasting. International Conference on Artificial Neural Networks. Springer, 2022, pp. 605–616.
172. Song, J.; Son, J.; Seo, D.h.; Han, K.; Kim, N.; Kim, S.W. ST-GAT: A Spatio-Temporal Graph Attention Network for Accurate Traffic Speed Prediction. Proceedings of the 31st ACM International Conference on Information & Knowledge Management, 2022, pp. 4500–4504.
173. Kim, D.; Cho, Y.; Kim, D.; Park, C.; Choo, J. Residual Correction in Real-Time Traffic Forecasting. Proceedings of the 31st ACM International Conference on Information & Knowledge Management, 2022, pp. 962–971.
174. Li, G.; Wang, X.; Njoo, G.S.; Zhong, S.; Chan, S.H.G.; Hung, C.C.; Peng, W.C. A Data-Driven Spatial-Temporal Graph Neural Network for Docked Bike Prediction. 2022 IEEE 38th International Conference on Data Engineering (ICDE). IEEE, 2022, pp. 713–726.
175. Shen, Y.; Li, L.; Xie, Q.; Li, X.; Xu, G. A Two-Tower Spatial-Temporal Graph Neural Network for Traffic Speed Prediction. Pacific-Asia Conference on Knowledge Discovery and Data Mining. Springer, 2022, pp. 406–418.
176. Sun, J.; Li, J.; Wu, C.; Tang, Z.; Wu, C. Ada-STNet: A Dynamic AdaBoost Spatio-Temporal Network for Traffic Flow Prediction. ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2022, pp. 5478–5482.
177. Shao, Z.; Zhang, Z.; Wei, W.; Wang, F.; Xu, Y.; Cao, X.; Jensen, C.S. Decoupled Dynamic Spatial-Temporal Graph Neural Network for Traffic Forecasting. *Proc. VLDB Endow.* **2022**, *15*, 2733–2746.
178. Lan, S.; Ma, Y.; Huang, W.; Wang, W.; Yang, H.; Li, P. DSTAGNN: Dynamic spatial-temporal aware graph neural network for traffic flow forecasting. International Conference on Machine Learning. PMLR, 2022, pp. 11906–11917.
179. Rao, X.; Wang, H.; Zhang, L.; Li, J.; Shang, S.; Han, P. FOGS: First-order gradient supervision with learning-based graph for traffic flow forecasting. Proceedings of International Joint Conference on Artificial Intelligence, IJCAI. ijcai. org, 2022.
180. Choi, J.; Choi, H.; Hwang, J.; Park, N. Graph neural controlled differential equations for traffic forecasting. Proceedings of the AAAI Conference on Artificial Intelligence, 2022, Vol. 36, pp. 6367–6374.
181. Zhang, C.; Zhang, S.; Yu, S.; James, J. Graph-Based Traffic Forecasting via Communication-Efficient Federated Learning. 2022 IEEE Wireless Communications and Networking Conference (WCNC). IEEE, 2022, pp. 2041–2046.
182. Liu, D.; Wang, J.; Shang, S.; Han, P. MSDR: Multi-step dependency relation networks for spatial temporal forecasting. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 1042–1050.
183. Lu, B.; Gan, X.; Zhang, W.; Yao, H.; Fu, L.; Wang, X. Spatio-Temporal Graph Few-Shot Learning with Cross-City Knowledge Transfer. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 1162–1172.
184. Li, P.; Fang, J.; Chao, P.; Zhao, P.; Liu, A.; Zhao, L. JS-STDGN: A Spatial-Temporal Dynamic Graph Network Using JS-Graph for Traffic Prediction. International Conference on Database Systems for Advanced Applications. Springer, 2022, pp. 191–206.
185. Shao, Z.; Zhang, Z.; Wang, F.; Xu, Y. Pre-training Enhanced Spatial-temporal Graph Neural Network for Multivariate Time Series Forecasting. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 1567–1577.
186. Yu, H.; Li, T.; Yu, W.; Li, J.; Huang, Y.; Wang, L.; Liu, A. Regularized Graph Structure Learning with Semantic Knowledge for Multi-variates Time-Series Forecasting. Proceedings of International Joint Conference on Artificial Intelligence, IJCAI. ijcai. org, 2022.
187. Tang, J.; Qian, T.; Liu, S.; Du, S.; Hu, J.; Li, T. Spatio-Temporal Latent Graph Structure Learning for Traffic Forecasting. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022.
188. Ji, J.; Wang, J.; Jiang, Z.; Jiang, J.; Zhang, H. STDEN: Towards Physics-guided Neural Networks for Traffic Flow Prediction. Proceedings of the AAAI Conference on Artificial Intelligence, 2022, Vol. 36, pp. 4048–4056.
189. Chen, Y.; Segovia-Dominguez, I.; Coskunuzer, B.; Gel, Y. TAMP-S2GCNets: coupling time-aware multipersistence knowledge representation with spatio-supra graph convolutional networks for time-series forecasting. International Conference on Learning Representations, 2022.
190. Xue, Y.; Fan, X.; Huang, Y.; Zhang, X.; Wang, R. Traffic Forecasting Model Based on Two-stage Stacked Graph Convolution Network. 2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD). IEEE, 2022, pp. 1089–1094.
191. Zhuang, D.; Wang, S.; Koutsopoulos, H.; Zhao, J. Uncertainty Quantification of Sparse Travel Demand Prediction with Spatial-Temporal Graph Neural Networks. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 4639–4647.
192. Lee, H.; Jin, S.; Chu, H.; Lim, H.; Ko, S. Learning to Remember Patterns: Pattern Matching Memory Networks for Traffic Forecasting. International Conference on Learning Representations, 2022.
193. Hermes, L.; Hammer, B.; Melnik, A.; Velioglu, R.; Vieth, M.; Schilling, M. A Graph-based U-Net Model for Predicting Traffic in unseen Cities. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022.
194. Feng, Y.; Han, F.; Zhao, S. A Graph Convolutional Stacked Temporal Attention Neural Network for Traffic Flow Forecasting. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022, pp. 1–7.

195. Li, S.; Ge, L.; Lin, Y.; Zeng, B. Adaptive Spatial-Temporal Fusion Graph Convolutional Networks for Traffic Flow Forecasting. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022, pp. 1–8.
196. Cao, S.; Wu, L.; Zhang, R.; Li, J.; Wu, D. Capturing Local and Global Spatial-Temporal Correlations of Spatial-Temporal Graph Data for Traffic Flow Prediction. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022, pp. 1–8.
197. Hu, J.; Lin, X.; Wang, C. MGCN: Dynamic Spatio-Temporal Multi-Graph Convolutional Neural Network. 2022 International Joint Conference on Neural Networks (IJCNN). IEEE, 2022, pp. 1–9.
198. Gao, Y.; Zhou, C.; Rong, J.; Wang, Y.; Liu, S. Short-Term Traffic Speed Forecasting Using a Deep Learning Method Based on Multitemporal Traffic Flow Volume. *IEEE Access* **2022**, *10*, 82384–82395.
199. Axenie, C.; Bortoli, S. Road traffic prediction dataset. **2020**. doi:10.5281/zenodo.3653880.
200. Hou, Y.; Chen, J.; Wen, S. The effect of the dataset on evaluating urban traffic prediction. *Alexandria Engineering Journal* **2021**, *60*, 597–613.
201. Braz, F.J.; Ferreira, J.; Gonçalves, F.; Weege, K.; Almeida, J.; Baldo, F.; Gonçalves, P. Road traffic forecast based on meteorological information through deep learning methods. *Sensors* **2022**, *22*, 4485.
202. Ma, H.; Zhou, M.; Ouyang, X.; Yin, D.; Jiang, R.; Song, X. Forecasting Regional Multimodal Transportation Demand with Graph Neural Networks: An Open Dataset. 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC). IEEE, 2022, pp. 3263–3268.
203. Prado-Rujas, I.I.; Serrano, E.; García-Dopico, A.; Córdoba, M.L.; Pérez, M.S. Combining heterogeneous data sources for spatio-temporal mobility demand forecasting. *Information Fusion* **2023**, *91*, 1–12.
204. Jiang, R.; Cai, Z.; Wang, Z.; Yang, C.; Fan, Z.; Chen, Q.; Tsubouchi, K.; Song, X.; Shibasaki, R. DeepCrowd: A deep model for large-scale citywide crowd density and flow prediction. *IEEE Transactions on Knowledge and Data Engineering* **2022**.
205. Wu, Z.; Zheng, D.; Pan, S.; Gan, Q.; Long, G.; Karypis, G. Traversenet: Unifying space and time in message passing for traffic forecasting. *IEEE Transactions on Neural Networks and Learning Systems* **2022**.
206. Xiao, Z.; Xiao, H.; Jiang, H.; Chen, W.; Chen, H.; Regan, A.C. Exploring human mobility patterns and travel behavior: A focus on private cars. *IEEE Intelligent Transportation Systems Magazine* **2021**, *14*, 129–146.
207. Xu, Z.; Tang, N.; Xu, C.; Cheng, X. Data science: connotation, methods, technologies, and development. *Data Science and Management* **2021**, *1*, 32–37.
208. Shao, Y.; Li, H.; Gu, X.; Yin, H.; Li, Y.; Miao, X.; Zhang, W.; Cui, B.; Chen, L. Distributed Graph Neural Network Training: A Survey. *arXiv preprint arXiv:2211.00216* **2022**.
209. Jiang, W.; He, M.; Gu, W. Internet Traffic Prediction with Distributed Multi-Agent Learning. *Applied System Innovation* **2022**, *5*, 121.
210. He, Q.; Dong, Z.; Chen, F.; Deng, S.; Liang, W.; Yang, Y. Pyramid: enabling hierarchical neural networks with edge computing. Proceedings of the ACM Web Conference 2022, 2022, pp. 1860–1870.
211. Wang, C.; Zhang, K.; Wang, H.; Chen, B. Auto-STGCN: Autonomous spatial-temporal graph convolutional network search based on reinforcement learning and existing research results. *arXiv preprint arXiv:2010.07474* **2020**.
212. Munikoti, S.; Agarwal, D.; Das, L.; Halappanavar, M.; Natarajan, B. Challenges and opportunities in deep reinforcement learning with graph neural networks: A comprehensive review of algorithms and applications. *arXiv preprint arXiv:2206.07922* **2022**.
213. Mingshuo, N.; Dongming, C.; Dongqi, W. Reinforcement Learning on Graph: A Survey. *arXiv preprint arXiv:2204.06127* **2022**.
214. Atwood, J.; Towsley, D. Diffusion-convolutional neural networks. *Advances in neural information processing systems* **2016**, *29*.